MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
4000 PROCESS PLANT: EMERGENCY SERVICES BUILDING
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Emergency Services Building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Emergency Services Building only.

4. DESIGN METHODOLOGY & ASSUMPTIONS

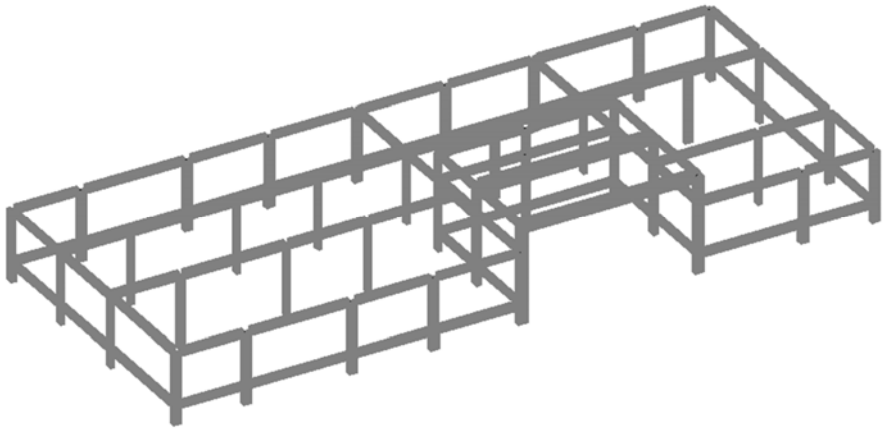
This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Building was modelled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.

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STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete.

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

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5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Roof Dead Loads

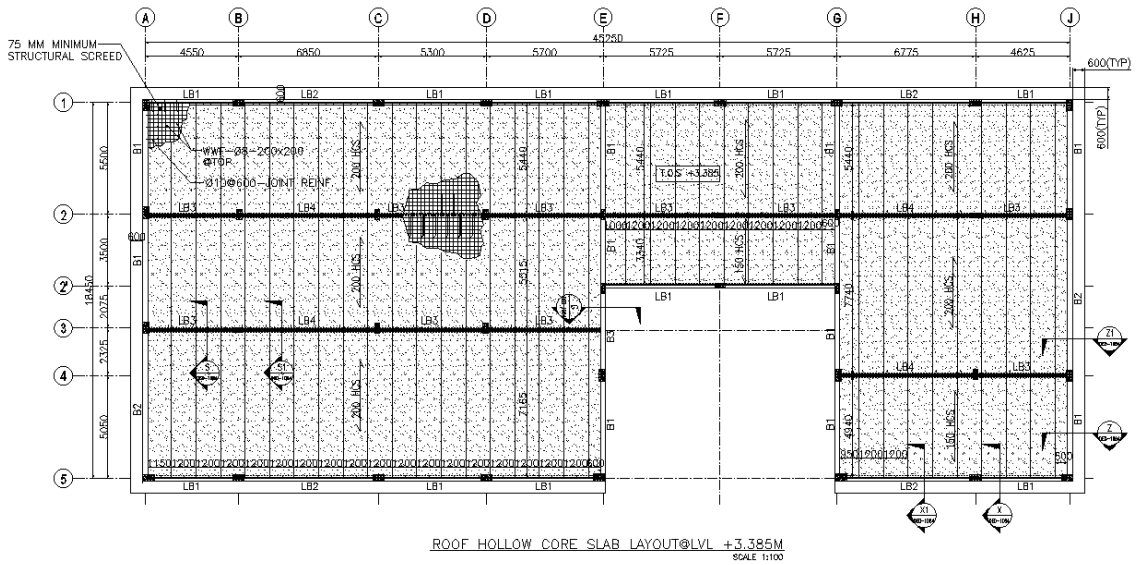
- Provision for Electrical Lighting = 0.05 kN/m²

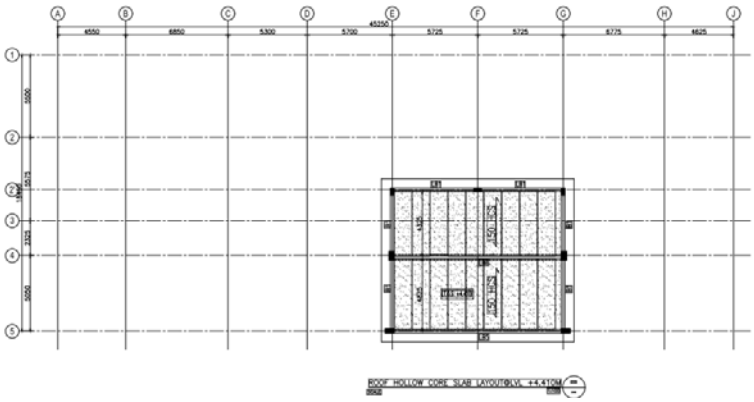
Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = **0.2 kN/m²**





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- Hollow Core Slab

@ LVL +3.385m

(HCS-200) b/w Grid Line's (1 & 2), (2 & 3), (4 & 5) = **3.10 KN/m2**

(HCS-150) b/w Grid Line's (2 & 2') = **2.52 KN/m2**

(HCS-200) b/w Grid Line's (2 & 4), (3 & 5) = **3.10 KN/m2**

@ LVL +5.125m

(HCS-150) b/w Grid Line's (2' & 5) = **2.52 KN/m2**

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

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Sub-total = **3.1 kN/ m²**

@ LVL +3.385m

Total DL b/w Grid Line's (1 & 2), (2 & 3), (4 & 5) = 3.1+3.1+0.2 = **6.40 kn/m2**

Total DL b/w Grid Line's (2 & 2') =3.1+0.2+2.52 = **5.82 KN/m2**

Total DL b/w Grid Line's (2 & 4), (3 & 5) =3.1+0.2+3.10 = **6.40 KN/m2**

@ LVL +5.125m

Total DL b/w Grid Line's (2' & 5) =3.1+0.2+2.52 = **5.82 KN/m2**

TOTAL DL as UDL ON BEAMS

@ LVL +3.385m

- UDL on LB's Along Grid Line-1 = 6.40 x 5.50/2 = **17.60 Kn/m**
- UDL on LB's Along Grid Line-5 b/w Grid A & E = 6.40 x 7.375/2 =**23.6 Kn/m**
- UDL on LB's Along Grid Line-5 b/w Grid G & J =6.40 x 5.05/2 =**16.10 Kn/m**
- UDL on LB's Along Grid Line-2' b/w Grid E & G =5.82 x 3.50/2 =**10.18 Kn/m**
- UDL on LB's Along Grid Line-3 b/w Grid A & E =24.82+(6.40 x 5.575/2) =**42.66 Kn/m**
- UDL on LB's Along Grid Line-2 b/w Grid A & E =17.6+(6.40 x 5.575/2) =**35.44 Kn/m**
- UDL on LB's Along Grid Line-2 b/w Grid E & G =17.6+(5.82 x 3.50/2) =**27.80 Kn/m**
- UDL on LB's Along Grid Line-2 b/w Grid G & J =17.6+(6.40 x 7.90/2) =**42.88 Kn/m**
- UDL on LB's Along Grid Line-4 b/w Grid G & J =16.1+(6.40 x 7.90/2) =**41.4 Kn/m**

@ LVL +5.125m

- UDL on LB's Along Grid Line-2 b/w Grid E & G =5.82 x 4.50/2=**13.10 Kn/m**
- UDL on LB's Along Grid Line-5 b/w Grid E & G =5.82 x 5.05/2=**14.70 Kn/m**
- UDL on LB's Along Grid Line-4 b/w Grid E & G =13.1+14.7=**27.80 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)
  - 16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**
  - 16 x 0.25 x (4) m clear height = **16 kN/m**

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5.2. Live Load

5.2.1. Roof Live Loads(Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on LB's Along Grid Line-1 = 0.50 x 5.50/2 = **1.375 Kn/m**
- UDL on LB's Along Grid Line-5 b/w Grid A & E = 0.50 x 7.375/2 =**1.85 Kn/m**
- UDL on LB's Along Grid Line-5 b/w Grid G & J =0.50 x 5.05/2 =**1.26 Kn/m**
- UDL on LB's Along Grid Line-2' b/w Grid E & G =0.50 x 3.50/2 =**0.875 Kn/m**
- UDL on LB's Along Grid Line-3 b/w Grid A & E =1.85+(0.50 x 5.575/2) =**3.25 Kn/m**
- UDL on LB's Along Grid Line-2 b/w Grid A & E =1.375+(0.50 x 5.575/2) =**2.77 Kn/m**
- UDL on LB's Along Grid Line-2 b/w Grid E & G =1.375+(0.50 x 3.50/2) =**2.25 Kn/m**
- UDL on LB's Along Grid Line-2 b/w Grid G & J =1.375+(0.50 x 7.90/2) =**3.35 Kn/m**
- UDL on LB's Along Grid Line-4 b/w Grid G & J =1.26+(0.50 x 7.90/2) =**3.23 Kn/m**

@ LVL +5.125m

- UDL on LB's Along Grid Line-2 b/w Grid E & G =0.50 x 4.50/2=**1.125 Kn/m**
- UDL on LB's Along Grid Line-5 b/w Grid E & G =0.50 x 5.05/2=**1.26 Kn/m**
- UDL on LB's Along Grid Line-4 b/w Grid E & G =1.125+1.25=**2.385 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

5.3. Wind Load(WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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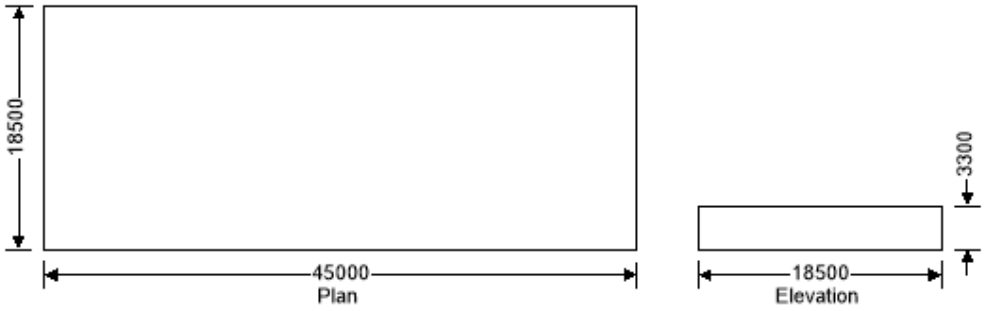
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5.4. Wind Load (Analytical Verification)

5.4.1. @3.3M LVL-WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = **45000** mm
Width of building;	d = **18500** mm
Height to eaves;	H = **3300** mm
Mean height;	h = **3300** mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)

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Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	148.54	24.88
B	3.30	-0.50	1.29	-1.26	148.50	-186.57
C	3.30	-0.70	1.29	-1.48	61.05	-90.07
D	3.30	-0.70	1.29	-1.48	61.05	-90.07

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	148.54	235.43

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Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
B	3.30	-0.50	1.29	0.16	148.50	23.92
C	3.30	-0.70	1.29	-0.06	61.05	-3.54
D	3.30	-0.70	1.29	-0.06	61.05	-3.54

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	61.07	10.23
B	3.30	-0.28	1.29	-1.01	61.05	-61.88
C	3.30	-0.70	1.29	-1.48	148.50	-219.10
D	3.30	-0.70	1.29	-1.48	148.50	-219.10

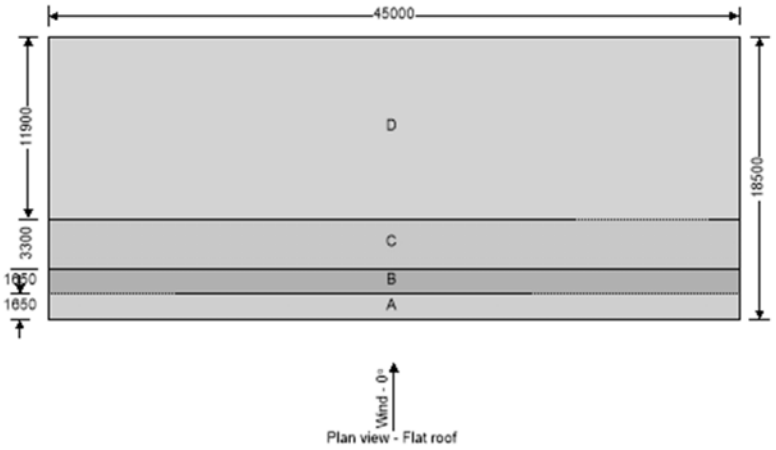
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

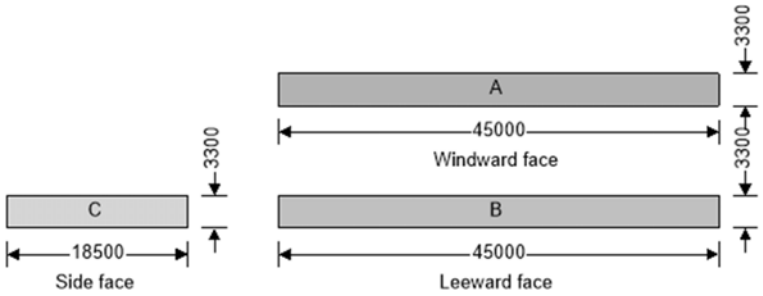
Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	61.07	96.79
B	3.30	-0.28	1.29	0.40	61.05	24.65
C	3.30	-0.70	1.29	-0.06	148.50	-8.61
D	3.30	-0.70	1.29	-0.06	148.50	-8.61

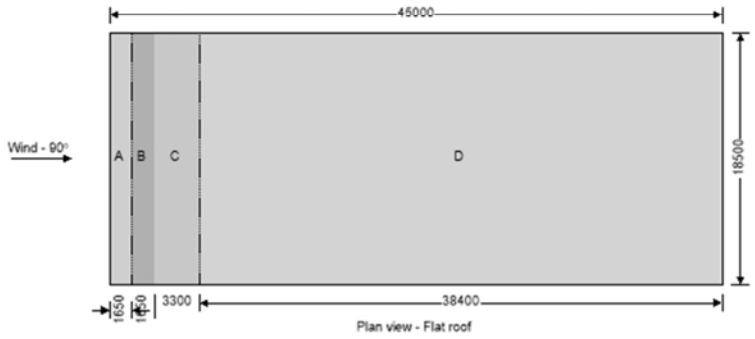
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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

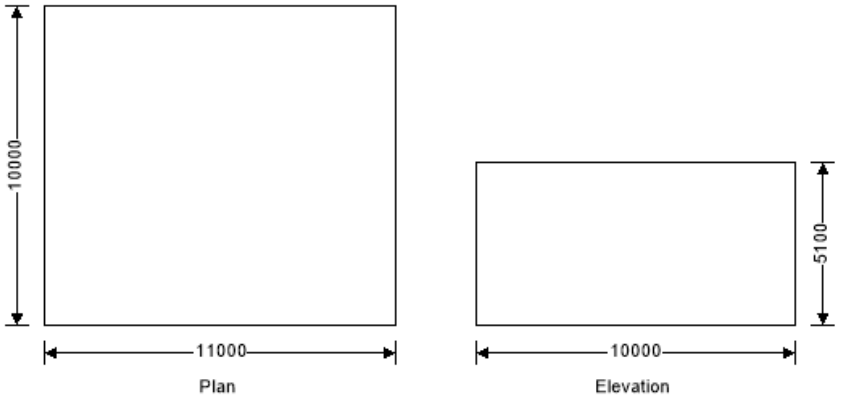
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5.4.2. @5.125M LVL-WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = **11000** mm
Width of building;	d = **10000** mm
Height to eaves;	H = 5100 mm
Mean height;	h = 5100 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

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Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (Table 27.3-1)	q_z (kN/m^2)
4572	1.03	1.29
5099	1.05	1.31

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.29	0.16	50.29	7.83
A_2	5.10	0.80	1.31	0.17	5.80	0.99
B	5.10	-0.50	1.31	-1.28	56.10	-71.67
C	5.10	-0.70	1.31	-1.50	51.00	-76.51
D	5.10	-0.70	1.31	-1.50	51.00	-76.51

Walls load case 2 - Wind 0, GC_{pi} -0.55, -1c_{pe}

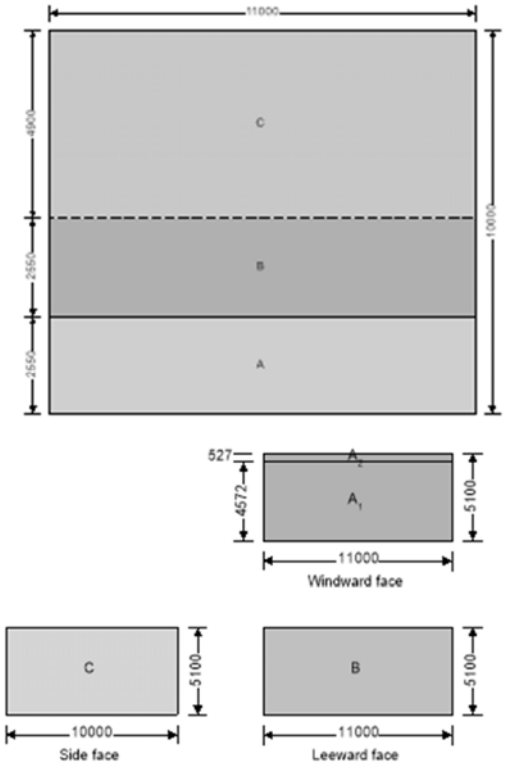
Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.29	1.60	50.29	80.31
A_2	5.10	0.80	1.31	1.61	5.80	9.35
B	5.10	-0.50	1.31	0.16	56.10	9.19
C	5.10	-0.70	1.31	-0.06	51.00	-3.01

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Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
D	5.10	-0.70	1.31	-0.06	51.00	-3.01



So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

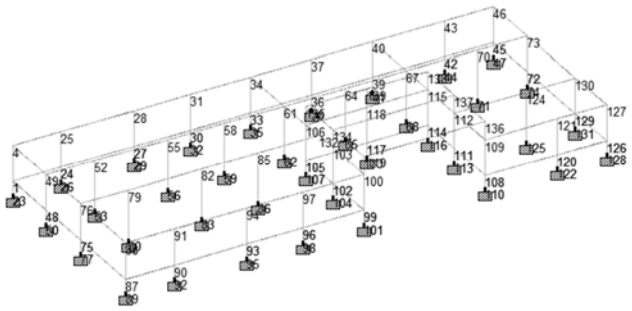
8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



EMERGENCY-5th MAY-20.std - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad
139	36	-0.036	-0.262	-26.088	26.089	-0.00
139	35	0.161	-0.437	-26.022	26.026	-0.00
139	21	0.415	-0.436	25.464	25.472	0.00
139	17	0.218	-0.262	25.398	25.401	0.00
118	36	-0.031	-0.223	-19.373	19.374	-0.00
118	35	0.153	-0.371	-19.234	19.238	-0.00
139	15	0.417	-0.455	19.147	19.157	0.00
118	21	0.379	-0.371	19.083	19.091	0.00
118	17	0.195	-0.222	18.944	18.946	0.00
139	20	3.224	-0.436	16.839	17.151	0.00
139	16	3.027	-0.262	16.773	17.046	0.00
94	20	0.018	-0.248	15.994	15.995	0.00
28	20	0.270	-0.193	-15.939	15.943	-0.00
94	16	0.298	-0.149	15.312	15.316	0.00
28	16	0.256	-0.116	-15.263	15.266	-0.00
118	15	0.387	-0.386	14.422	14.432	0.00
94	36	-0.595	-0.150	-13.974	13.988	-0.00

Maximum Attained Displacement as per Analysis is 27mm<<<H/100=3300/100=33mm.

Hence Structure is safe against Drift check.

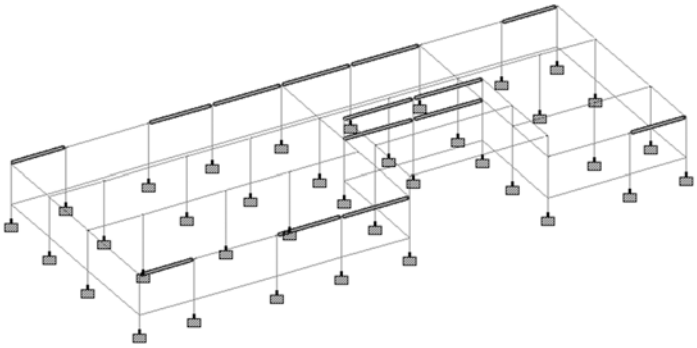
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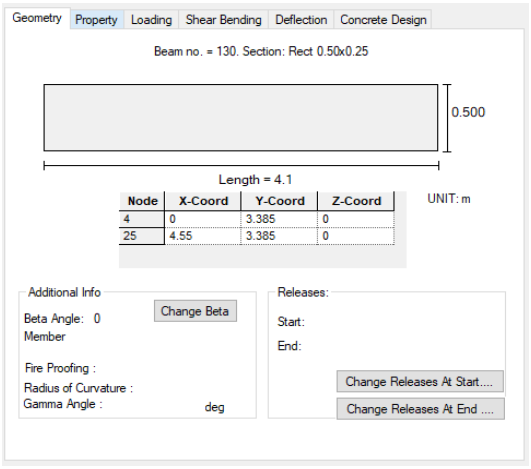
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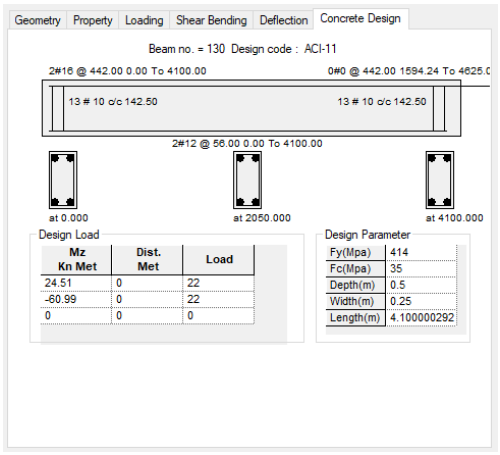
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 60 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **440 mm²**

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Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **25 Knm (Sagging)**

Ast2.req = **200mm²**

Hence Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bottom

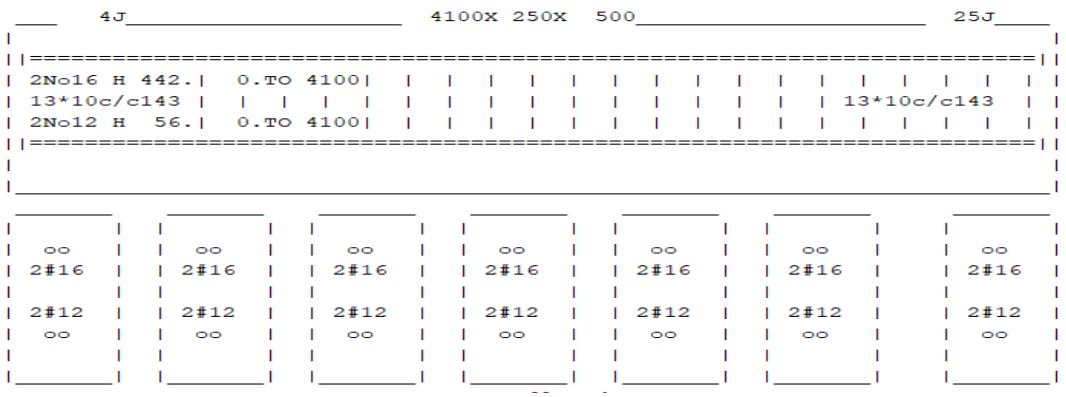
Ast1.pro= **602mm² >>>>>>(Ast.min, Ast1.req)**
Ast2.pro= **602mm² >>>>>>(Ast.min, Ast2.req)**

9.1.1. Staad Output For LB1

```
            B E A M   N O.   130 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   30.00 KNS  Vc=  195.02 KNS  Vs=    0.00 KNS
 Tu=   12.86 KN-MET  Tc=    4.0 KN-MET  Ts=   17.1 KN-MET  LOAD    27
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1610. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.24 SQ.CM.

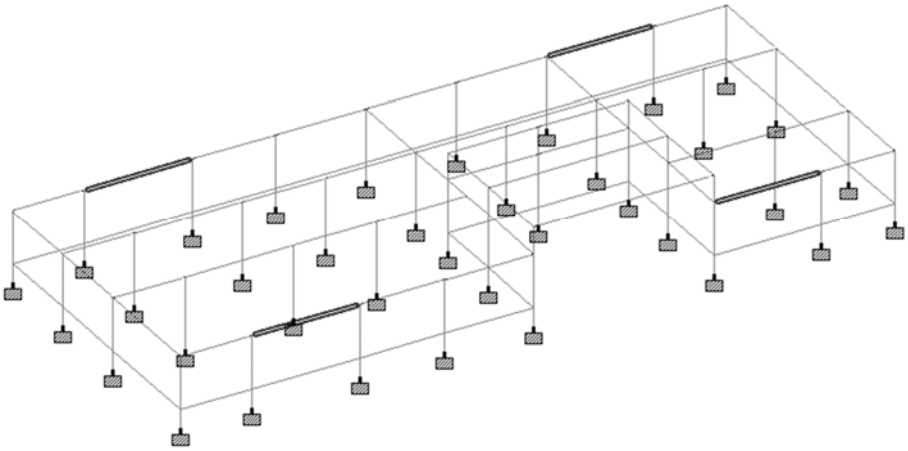
 AT END   SUPPORT - Vu=   50.66 KNS  Vc=  109.06 KNS  Vs=    0.00 KNS
 Tu=   10.26 KN-MET  Tc=    3.8 KN-MET  Ts=   13.7 KN-MET  LOAD    28
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1610. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   3.38 SQ.CM.
```

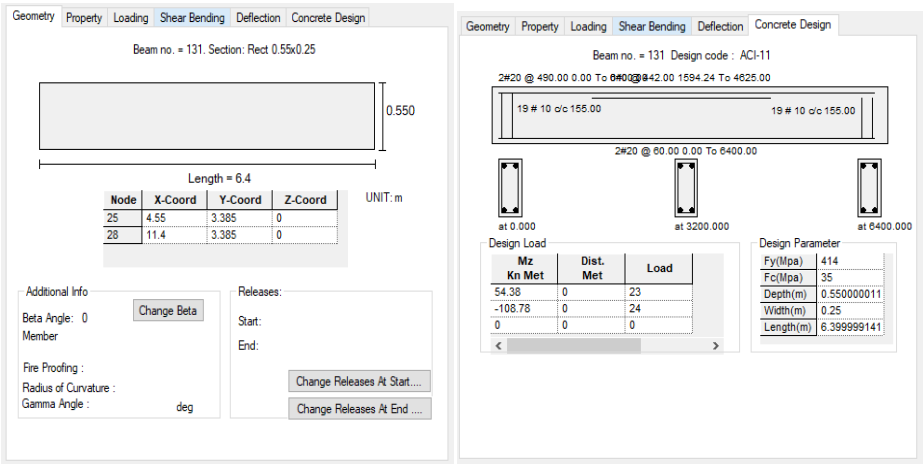


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9.2. Design of Ledger Beam(LB2)





Design of Ledger Beam (LB2)

However Provide 3-20 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom,..for Beams LB2 considering Long term Deflection check in Section 9.2.1

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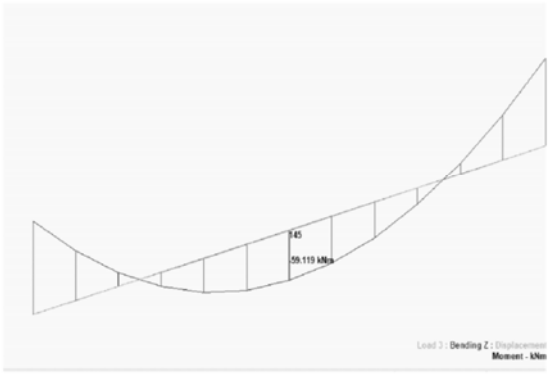
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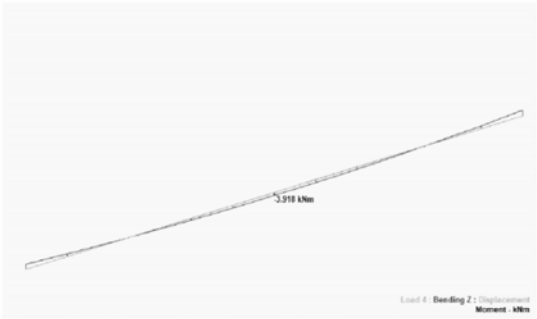
9.2.1. Long Term Deflection Check For LB2

(LB2-BEAM-LONGTERM DEFLECTION CHECK)

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	550	mm
width of beam, b	=	250	mm
Length of beam, L	=	6.80	m
Bottom bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of bottom bars	=	3.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	490.00	mm
(FROM STAAD RESULTS) Mdead	=	59.00	kN-m
Mlive	=	6.00	kN-m





CHECK DEFLECTION: $M_{d+\ell}$ = 65.00 kN-m

$$M_{sus} = M_d + 0.50M_\ell = 62.00 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$fr = 0.62\lambda\sqrt{fc'} = 3.67 \text{ Mpa}$$

$$n_s = \frac{Es}{Ec} = 7.19 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

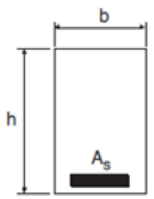
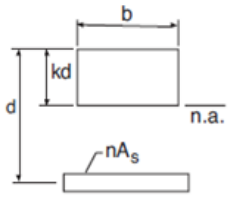
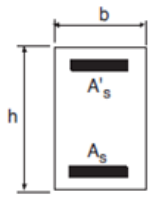
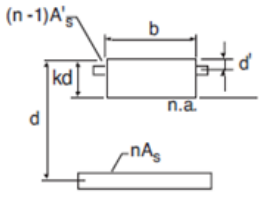
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

Page 2

$$I_g = \frac{bh^3}{12} = 3.47E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.04 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 138.14 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.06E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 3.27$$

effective moment of inertia

$$M_{cr} = \frac{ff'I_g}{y_t} = 46.23 \text{ kN-m}$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.78$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 2.22E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.75$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 2.06E+09 \text{ mm}^4 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

under dead + live load

$$\frac{M_{cr}}{M_{d+\lambda}} = 0.71$$

$$(I_e)_{d+\lambda} = \left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3\right] I_{cr} = 1.93E+09 \text{ mm}^4 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\lambda^2}{Ec(I_e)_d} = 4.610 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\lambda^2}{Ec(I_e)_{sus}} = 5.221 \text{ mm}$$

$$(\Delta_i)_{d+\lambda} = \frac{K(5/48)M_{d+\lambda}\lambda^2}{Ec(I_e)_{d+\lambda}} = 5.849 \text{ mm}$$

$$(\Delta_i)_\lambda = (\Delta_i)_{d+\lambda} - (\Delta_i)_d = 1.239 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\lambda \le \frac{\lambda}{180}$ = 37.78 mm > 1.24 mm *ok!*

$(\Delta_i)_\lambda \le \frac{\lambda}{360}$ = 18.89 mm > 1.24 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$\rho = \frac{As}{bd}$ = 0.0077

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.44 ←*ACI(318M-08) Sec. 9.5.2.5*

Page 4

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 7.54 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda$ = 8.78 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480}$ = 14.17 mm > 8.78 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240}$ = 28.33 mm > 8.78 mm *ok!*

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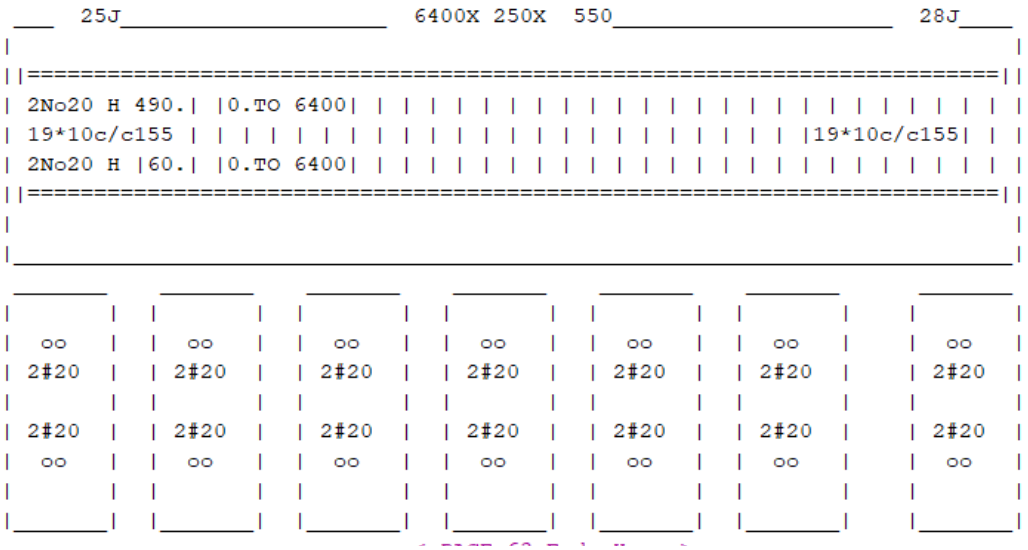
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9.2.2. Staad Output For LB2

```
                B E A M  N O.   131 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   68.91 KNS  Vc=  120.52 KNS  Vs=    0.00 KNS
 Tu=    7.57 KN-MET  Tc=    4.6 KN-MET  Ts=   10.1 KN-MET  LOAD    27
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 155. MM  C/C FOR 2710. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.42 SQ.CM.

 AT END   SUPPORT - Vu=   69.70 KNS  Vc=  119.75 KNS  Vs=    0.00 KNS
 Tu=    6.79 KN-MET  Tc=    4.5 KN-MET  Ts=    9.1 KN-MET  LOAD    26
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 155. MM  C/C FOR 2710. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.17 SQ.CM.
```

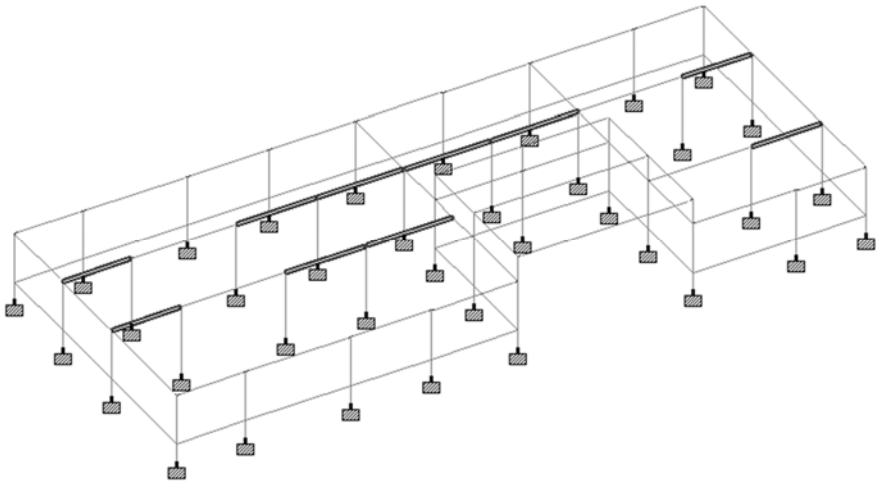


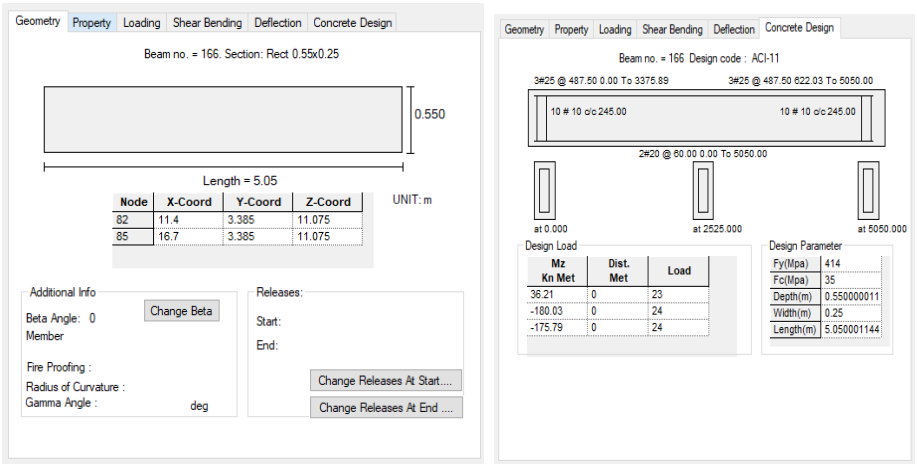
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9.3. Design of Ledger Beam(LB3)





Design of Ledger Beam (LB3)

However Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom,..for Beams LB3 considering Long term Deflection check in Section 9.3.1

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9.3.1. Long Term Deflection Check For LB3

(LB3-BEAM-LONGTERM DEFLECTION CHECK)

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	550	mm
width of beam, b	=	250	mm
Length of beam, L	=	5.70	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	3.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	487.50	mm
(FROM STAAD RESULTS) Mdead	=	98.00	kN-m
Mlive	=	7.00	kN-m







CHECK DEFLECTION: M_{d+l} = 105.00 kN-m

$$M_{sus} = M_d + 0.50 M_l = 101.50 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$f_r = 0.62\lambda\sqrt{f'_c} = 3.67 \text{ Mpa}$$

$$n_s = \frac{Es}{Ec} = 7.19 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

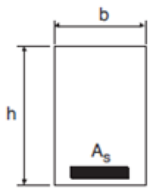
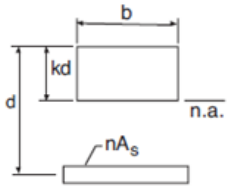
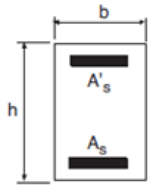
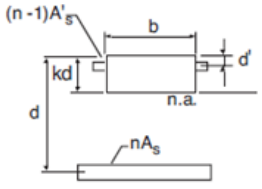
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

Page 2

$$I_g = \frac{bh^3}{12} = 3.47E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.02 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 165.25 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.48E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 2.35$$

effective moment of inertia

$$M_{cr} = \frac{fr \cdot I_g}{y_t} = 46.23 \text{ kN-m}$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead load

$$\frac{M_{cr}}{M_d} = 0.47$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.68E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.46$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.66E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\lambda}} = 0.44$$

$$(I_e)_{d+\lambda} = \left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3\right] I_{cr} = 1.65E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\lambda^2}{Ec(I_e)_d} = 7.079 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\lambda^2}{Ec(I_e)_{sus}} = 7.424 \text{ mm}$$

$$(\Delta_i)_{d+\lambda} = \frac{K(5/48)M_{d+\lambda}\lambda^2}{Ec(I_e)_{d+\lambda}} = 7.765 \text{ mm}$$

$$(\Delta_i)_\lambda = (\Delta_i)_{d+\lambda} - (\Delta_i)_d = 0.686 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\lambda \le \frac{\lambda}{180}$ = 31.67 mm > 0.69 mm *ok!*

$(\Delta_i)_\lambda \le \frac{\lambda}{360}$ = 15.83 mm > 0.69 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$\rho = \frac{As}{bd}$ = 0.0121

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.25 ←*ACI(318M-08) Sec. 9.5.2.5*

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 9.26 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda$ = 9.94 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480}$ = 11.88 mm > 9.94 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240}$ = 23.75 mm > 9.94 mm *ok!*

Page 4

9.3.2. Staad Output For LB3

```
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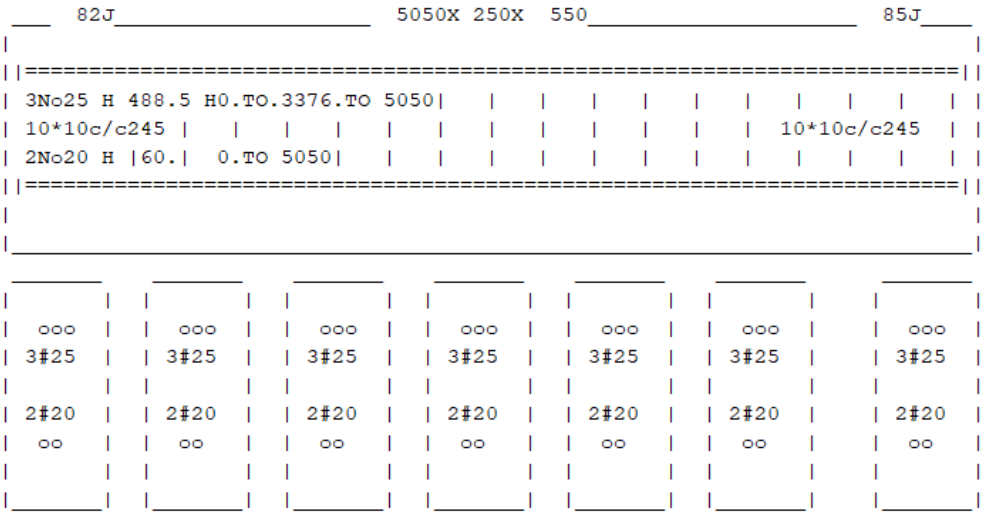
AT START SUPPORT - Vu=  137.53 KNS  Vc=  117.45 KNS  Vs=   65.93 KNS
Tu=    0.33 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD    23
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 2035. MM

AT END   SUPPORT - Vu=  135.85 KNS  Vc=  117.52 KNS  Vs=   63.61 KNS
Tu=    0.33 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD    23
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 2035. MM
```

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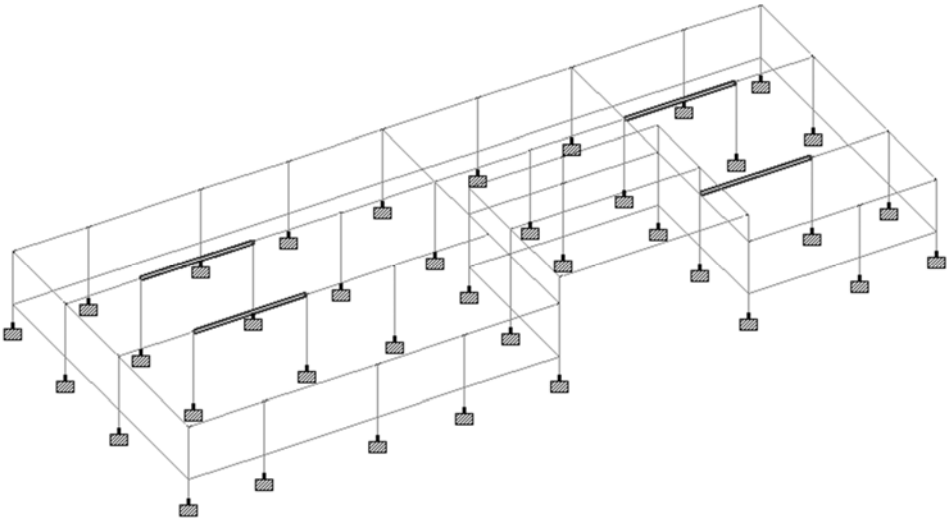


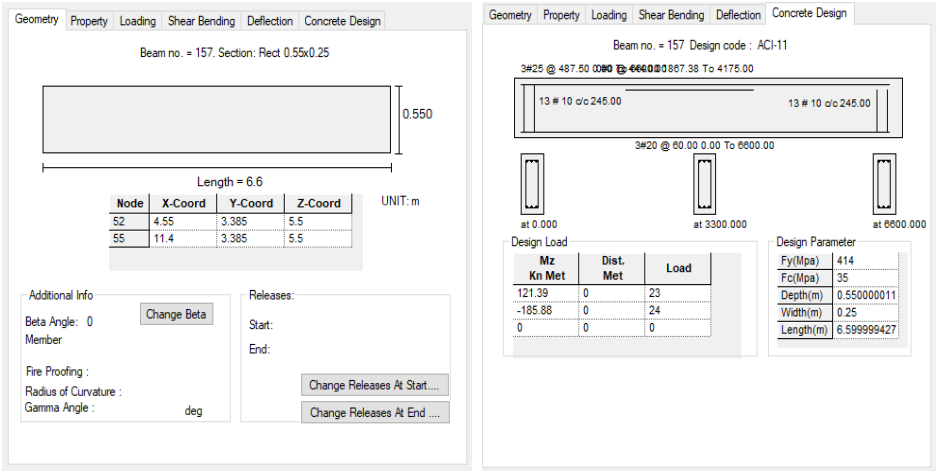
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9.4. Design of Ledger Beam(LB4)





Design of Ledger Beam (LB4)

However Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom,..for Beams LB4 considering Long term Deflection check in Section 9.4.1

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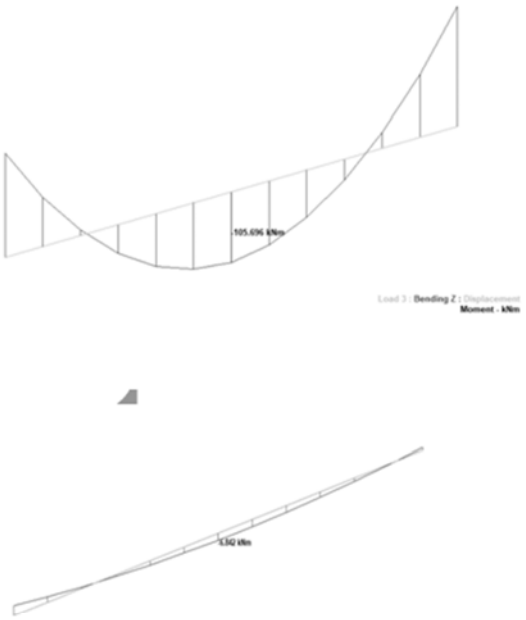
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9.4.1. Long Term Deflection Check For LB4

(LB4-BEAM-LONGTERM DEFLECTION CHECK)
INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	550	mm
width of beam, b	=	250	mm
Length of beam, L	=	6.80	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	5.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	487.50	mm
(FROM STAAD RESULTS) Mdead	=	105.00	kN-m
Mlive	=	8.00	kN-m



CHECK DEFLECTION: M_{d+L} = 113.00 kN-m

$M_{sus} = M_d + 0.50 M_L$ = 109.00 kN-m

modulus of rupture, modulus of elasticity, modular ratio

$f_r = 0.62\lambda\sqrt{f_c'}$ = 3.67 Mpa

$n_c = \frac{Es}{Ec}$ = 7.19 ←*ACI(318M-08) Sec. 9.5.2.3*

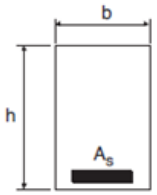
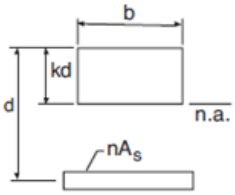
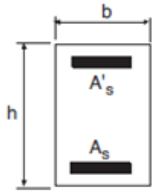
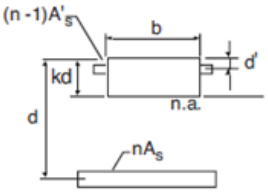
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

Page 2

$$I_g = \frac{bh^3}{12} = 3.47E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.01 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 201.11 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 2.13E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 1.63$$

effective moment of inertia

$$M_{cr} = \frac{fr\,I_g}{y_t} = 46.23 \text{ kN-m} \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

under dead load

$$\frac{M_{cr}}{M_d} = 0.44$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 2.24E+09 \text{ mm}^4 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.42$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 2.23E{+}09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.41$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 2.22E{+}09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d \ell^2}{Ec(I_e)_d} = 8.119 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus} \ell^2}{Ec(I_e)_{sus}} = 8.475 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell} \ell^2}{Ec(I_e)_{d+\ell}} = 8.827 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 0.708 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\ell \le \frac{\ell}{180}$ = 37.78 mm > 0.71 mm *ok!*

$(\Delta_i)_\ell \le \frac{\ell}{360}$ = 18.89 mm > 0.71 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months..... 1.4
6 months..... 1.2
3 months..... 1.0

$\rho = \frac{As}{bd}$ = 0.0201

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.00 ←*ACI(318M-08) Sec. 9.5.2.5*

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 8.45 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell$ = 9.15 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480}$ = 14.17 mm > 9.15 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240}$ = 28.33 mm > 9.15 mm *ok!*

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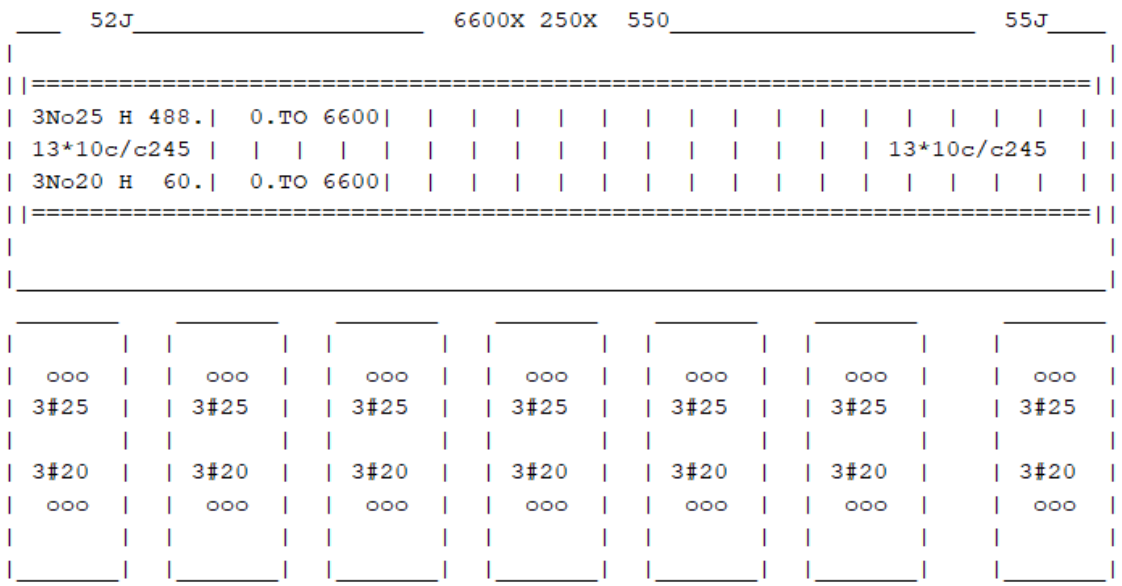
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9.4.2. Staad Output For LB4

```
        B E A M  N O.   157 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=  157.55 KNS  Vc=  123.78 KNS  Vs=   86.28 KNS
Tu=    0.02 KN-MET  Tc=    4.5 KN-MET  Ts=    0.0 KN-MET  LOAD    23
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 2810. MM

AT END   SUPPORT - Vu=  154.63 KNS  Vc=  124.45 KNS  Vs=   81.72 KNS
Tu=    0.02 KN-MET  Tc=    4.5 KN-MET  Ts=    0.0 KN-MET  LOAD    23
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 2810. MM
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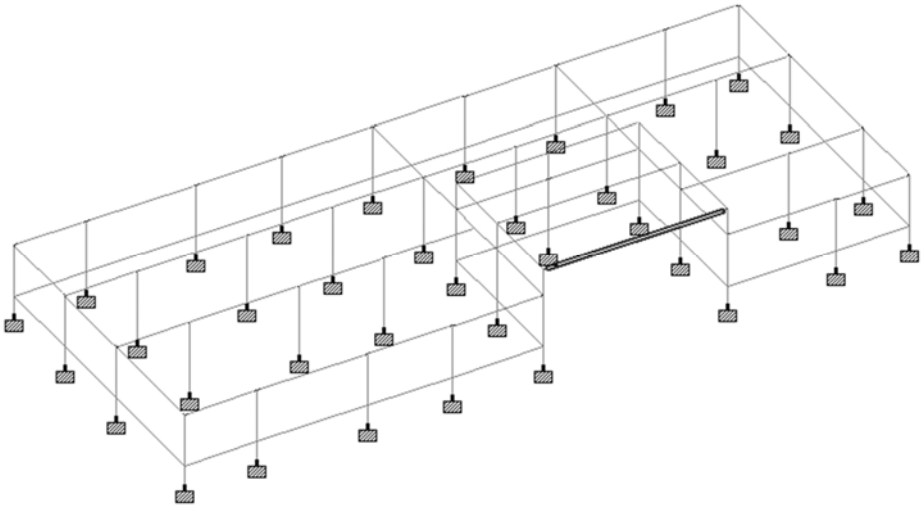


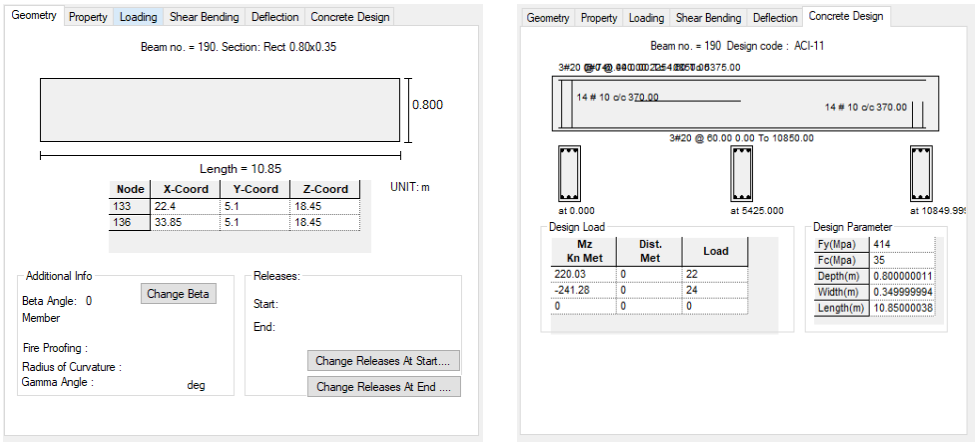
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9.5. Design of Ledger Beam(LB5)





Design of Ledger Beam (LB5)

However Provide 3-20 dia-Full Straight Bars, 2-20 dia Extra Rein@ Top &
5-20 dia-Full Straight Bottom,..for Beams LB5 considering Long term Deflection check in Section 9.5.1

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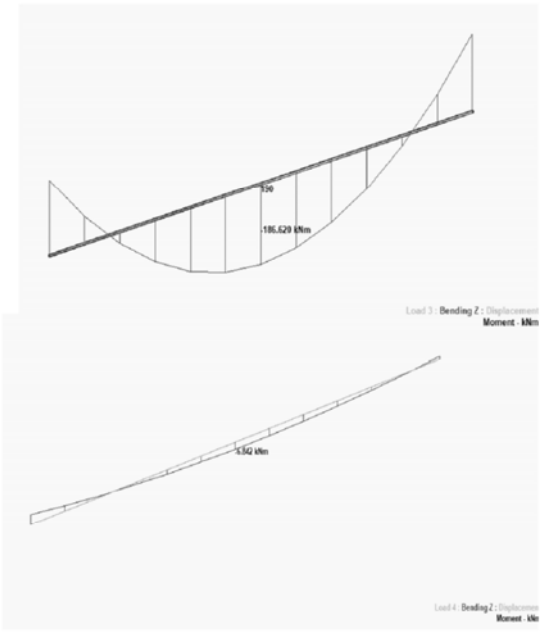
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9.5.1. Long Term Deflection Check For LB5

(LB5-BEAM-LONGTERM DEFLECTION CHECK)
INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	800	mm
width of beam, b	=	350	mm
Length of beam, L	=	11.00	m
Bottom bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of bottom bars	=	5.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	5.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	740.00	mm
(FROM STAAD RESULTS) Mdead	=	140.00	kN-m
Mlive	=	8.00	kN-m



CHECK DEFLECTION: $M_{d+\ell}$ = 148.00 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 144.00 kN-m

modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa

$n_s = \frac{Es}{Ec}$ = 7.19 *?ACI(318M-08) Sec. 9.5.2.3*

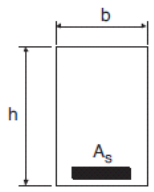
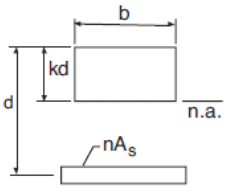
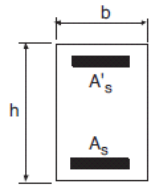
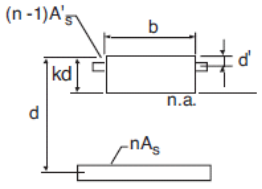
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 1.49E+10 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.03 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 188.67 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 4.22E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 3.54$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 136.94 \text{ kN-m}$$ *?ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.98$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.42E+10 \text{ mm}^4$$ *?ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.95$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.34E+10 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.93$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 1.27E+10 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 4.455 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 4.859 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 5.280 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 0.825 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\lambda \le \frac{\lambda}{180}$ = 61.11 mm > 0.83 mm *ok!*

$(\Delta_i)_\lambda \le \frac{\lambda}{360}$ = 30.56 mm > 0.83 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$\rho = \frac{As}{bd}$ = 0.0061

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.53 ←*ACI(318M-08) Sec. 9.5.2.5*

Page 4

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 7.46 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda$ = 8.28 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480}$ = 22.92 mm > 8.28 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240}$ = 45.83 mm > 8.28 mm *ok!*

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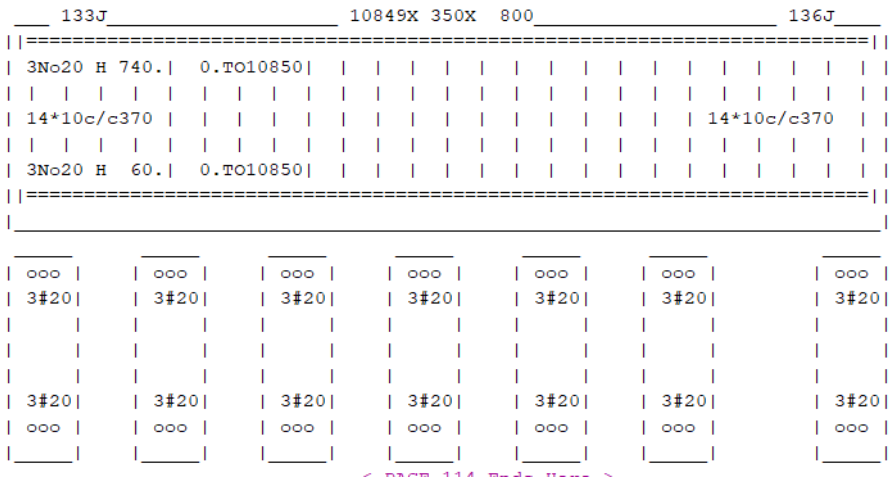
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9.5.2. Staad Output For LB5

```
            B E A M   N O.   190 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=    99.32 KNS   Vc=  261.51 KNS   Vs=     0.00 KNS
Tu=     4.75 KN-MET   Tc=    13.2 KN-MET   Ts=     0.0 KN-MET   LOAD      3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 370. MM  C/C FOR 4685. MM

AT END    SUPPORT - Vu=   100.37 KNS   Vc=  260.18 KNS   Vs=     0.00 KNS
Tu=     4.62 KN-MET   Tc=    13.2 KN-MET   Ts=     0.0 KN-MET   LOAD      3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 370. MM  C/C FOR 4685. MM
```

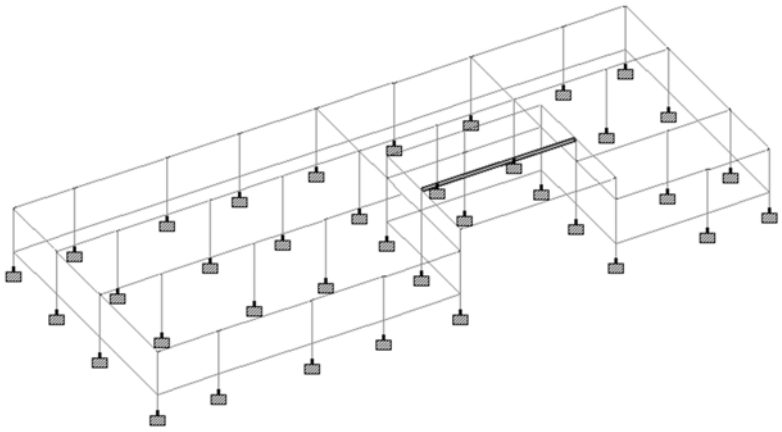


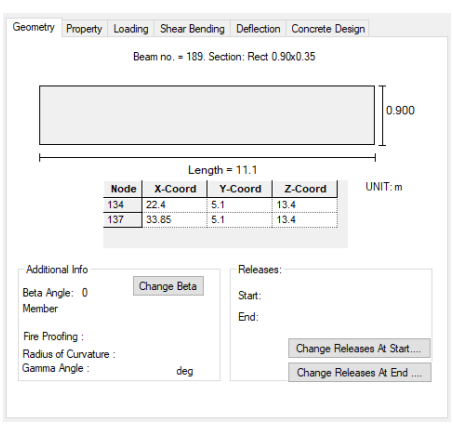
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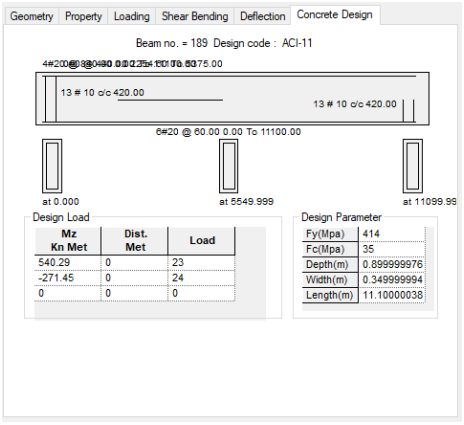
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9.6. Design of Ledger Beam(LB6)







Design of Ledger Beam (LB6)

However Provide 4-20 dia-Full Straight Bars, 2-20 dia Extra Rein@ Top &
5-25 dia-Full Straight, 3-25 dia Extra Rein@ Bottom,..for Beams LB6 considering Long term Deflection check in Section 9.6.1

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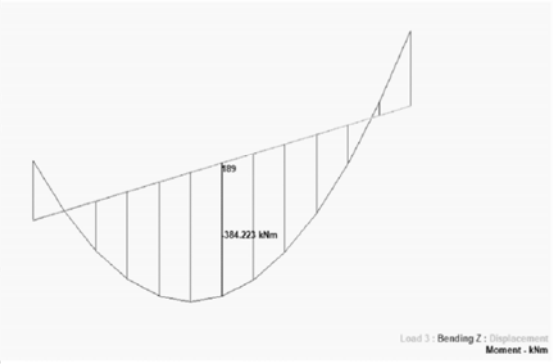
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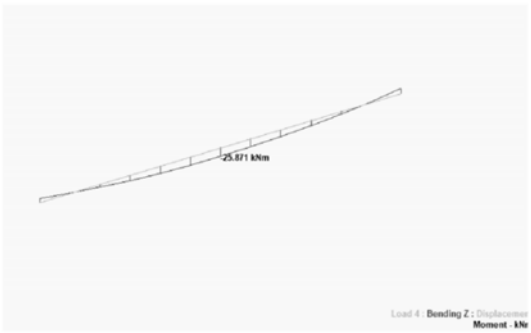
9.6.1. Long Term Deflection Check For LB6

(LB6-BEAM-LONGTERM DEFLECTION CHECK)

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/n
Unit wt. of steel	=	77.00	kN/n
Depth of beam, h	=	900	mm
width of beam, b	=	400	mm
Length of beam, L	=	11.00	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	8.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	6.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	837.50	mm
(FROM STAAD RESULTS) Mdead	=	384.00	kN-m
Mlive	=	25.00	kN-m





CHECK DEFLECTION: $M_{d+\ell}$ = 409.00 kN-m

$$M_{sus} = M_d + 0.50M_\ell \quad = \quad 396.50 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$fr = 0.62\lambda\sqrt{fc'} \quad = \quad 3.67 \text{ Mpa}$$

$$n_s = \frac{Es}{Ec} \quad = \quad 7.19 \qquad \text{?ACI(318M-08) Sec. 9.5.2.3}$$

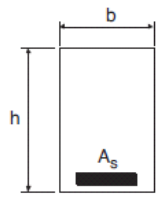
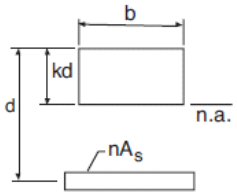
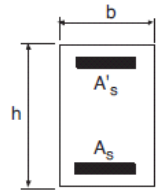
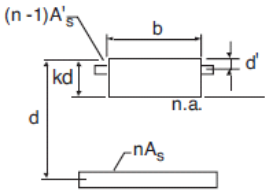
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 2.43E+10 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.01 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 280.48 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.17E+10 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 2.08$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 198.07 \text{ kN-m}$$ *?ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.52$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.34E+10 \text{ mm}^4$$ *?ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.50$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.33E+10 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\lambda}} = 0.48$$

$$(I_e)_{d+\lambda} = \left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3\right] I_{cr} = 1.31E+10 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$$K = 1.00$$

$$(\Delta_i)_d = \frac{K(5/48)M_d\lambda^2}{Ec(I_e)_d} = 12.957 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\lambda^2}{Ec(I_e)_{sus}} = 13.538 \text{ mm}$$

$$(\Delta_i)_{d+\lambda} = \frac{K(5/48)M_{d+\lambda}\lambda^2}{Ec(I_e)_{d+\lambda}} = 14.113 \text{ mm}$$

$$(\Delta_i)_\lambda = (\Delta_i)_{d+\lambda} - (\Delta_i)_d = 1.157 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\ell \le \frac{\ell}{180}$ = 61.11 mm > 1.16 mm *ok!*

$(\Delta_i)_\ell \le \frac{\ell}{360}$ = 30.56 mm > 1.16 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$\rho = \frac{As}{bd}$ = 0.0117

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.26 ←*ACI(318M-08) Sec. 9.5.2.5*

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 17.07 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell$ = 18.23 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480}$ = 22.92 mm > 18.23 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240}$ = 45.83 mm > 18.23 mm *ok!*

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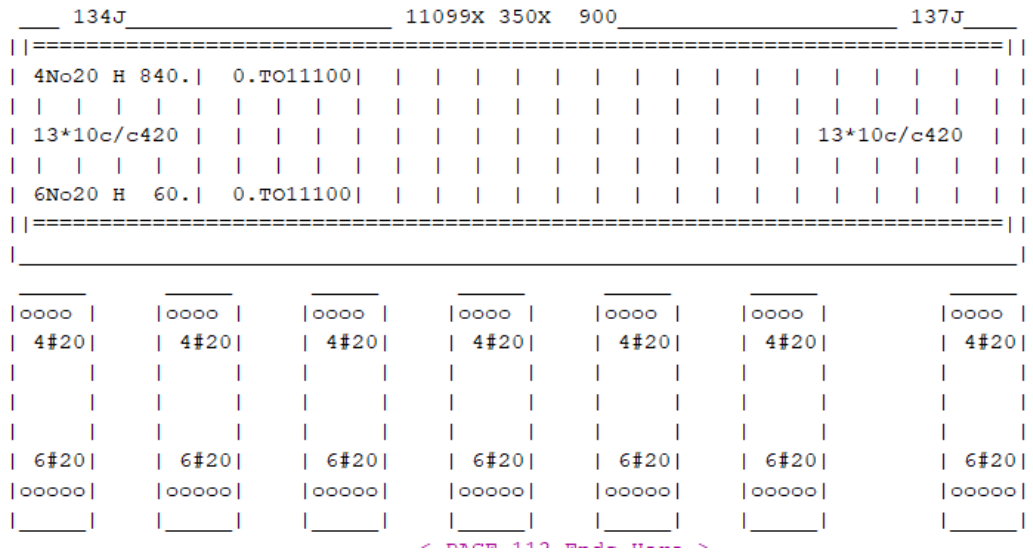
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9.6.2. Staad Output For LB6

```
        B E A M  N O.   189 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=  226.54 KNS  Vc=  518.56 KNS  Vs=    0.00 KNS
 Tu=    0.20 KN-MET  Tc=   15.3 KN-MET  Ts=    0.0 KN-MET  LOAD    22
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 420. MM  C/C FOR 4710. MM

 AT END   SUPPORT - Vu=  180.83 KNS  Vc=  448.22 KNS  Vs=    0.00 KNS
 Tu=    0.14 KN-MET  Tc=   15.1 KN-MET  Ts=    0.0 KN-MET  LOAD    14
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 420. MM  C/C FOR 4710. MM
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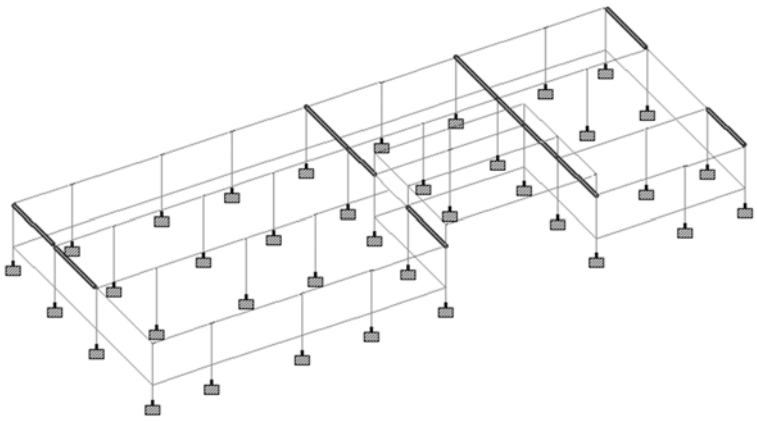


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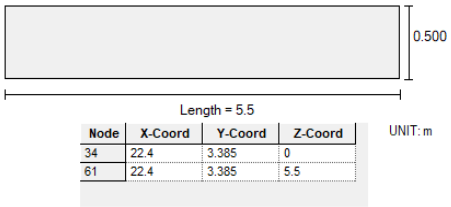
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9.7. Design of Beam-(B1)



Geometry | Property | Loading | Shear Bending | Deflection | Concrete Design

Beam no. = 171. Section: Rect 0.50x0.25



Node	X-Coord	Y-Coord	Z-Coord
34	22.4	3.385	0
61	22.4	3.385	5.5

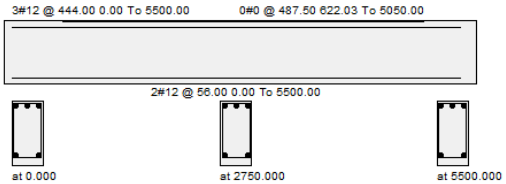
UNIT: m

Additional Info
Beta Angle: 0
Member
Fire Proofing :
Radius of Curvature :
Gamma Angle : deg
Change Beta

Releases:
Start:
End:
Change Releases At Start....
Change Releases At End ....

Geometry | Property | Loading | Shear Bending | Deflection | Concrete Design

Beam no. = 171 Design code : ACI-11



Design Load

Mz Kn Met	Dist. Met	Load
13.2	458.3	32
-29.76	0	38
0	0	0

Design Parameter

Fy(Mpa)	414
Fc(Mpa)	35
Depth(m)	0.5
Width(m)	0.25
Length(m)	5.5

Provide 3-16 dia-Full Straight Bars Top & Bottom

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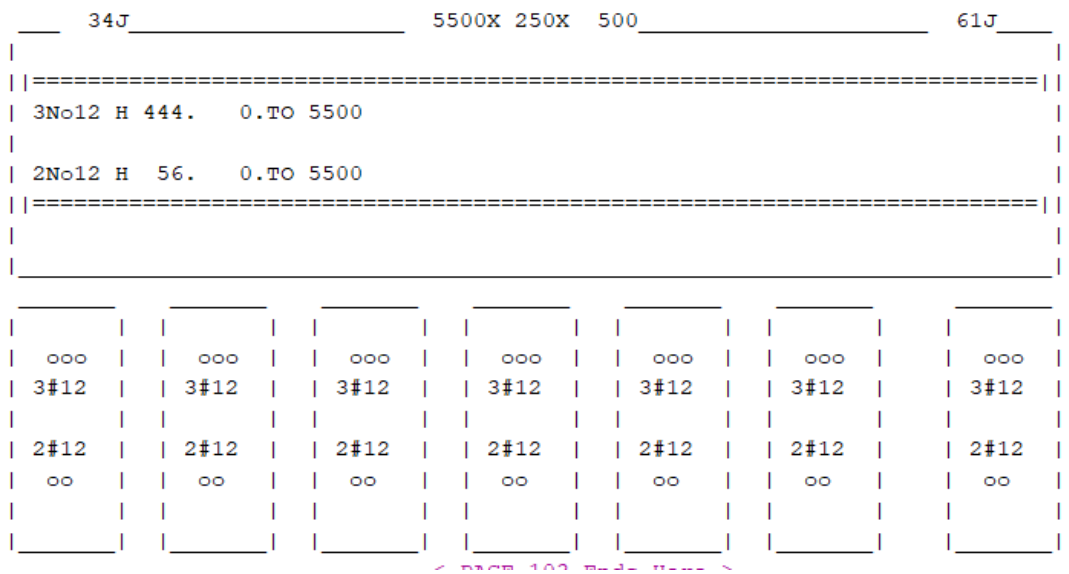
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9.7.1. Staad Output For B1

```
                B E A M  N O.   171 D E S I G N  R E S U L T S - SHEAR


      AT START SUPPORT - Vu=    7.35 KNS   Vc=  108.98 KNS   Vs=    0.00 KNS
      Tu=    0.54 KN-MET   Tc=    3.8 KN-MET   Ts=    0.0 KN-MET   LOAD     3
                       STIRRUPS ARE NOT REQUIRED.
      AT END   SUPPORT - Vu=    6.29 KNS   Vc=  108.98 KNS   Vs=    0.00 KNS
      Tu=    0.54 KN-MET   Tc=    3.8 KN-MET   Ts=    0.0 KN-MET   LOAD     3
                       STIRRUPS ARE NOT REQUIRED.
```

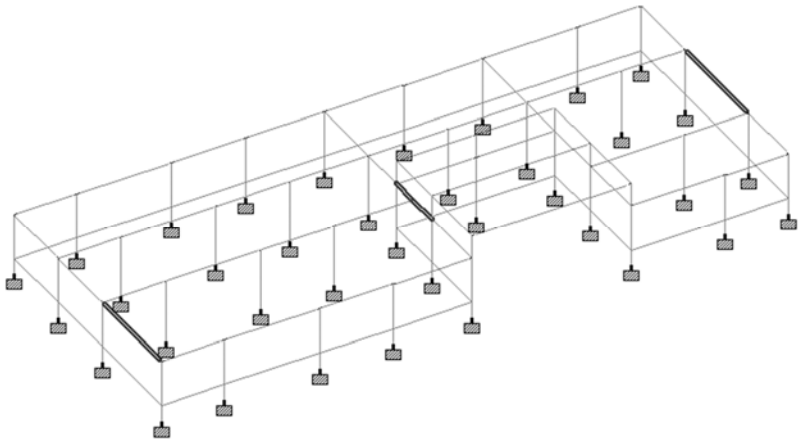


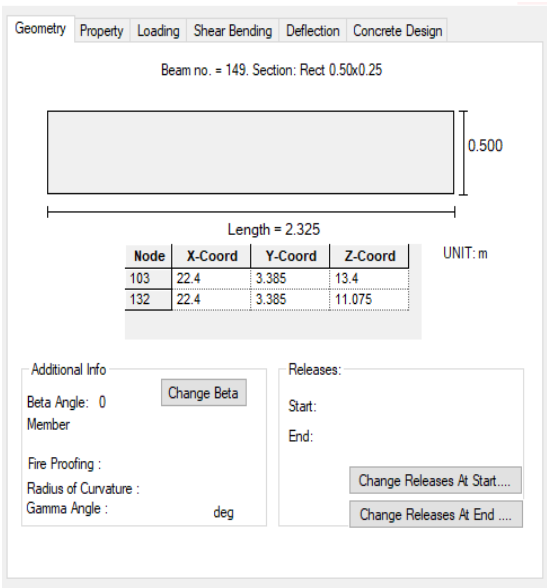
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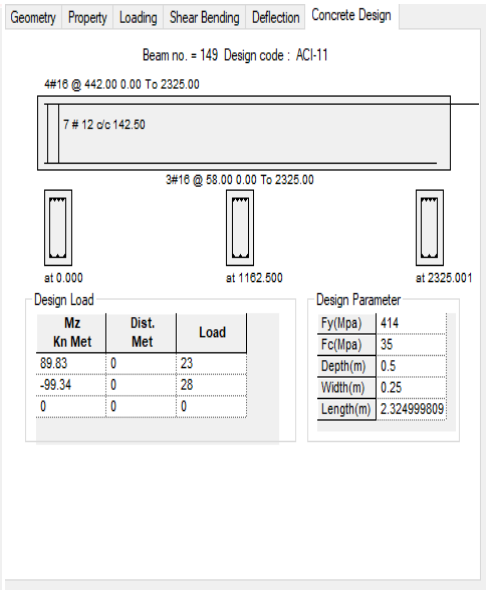
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9.8. Design of Beam-(B2)







Provide Equivalent 3-16 dia-Full Straight Bars & 2-16 Dia-Extra Reinf both Top & Bottom

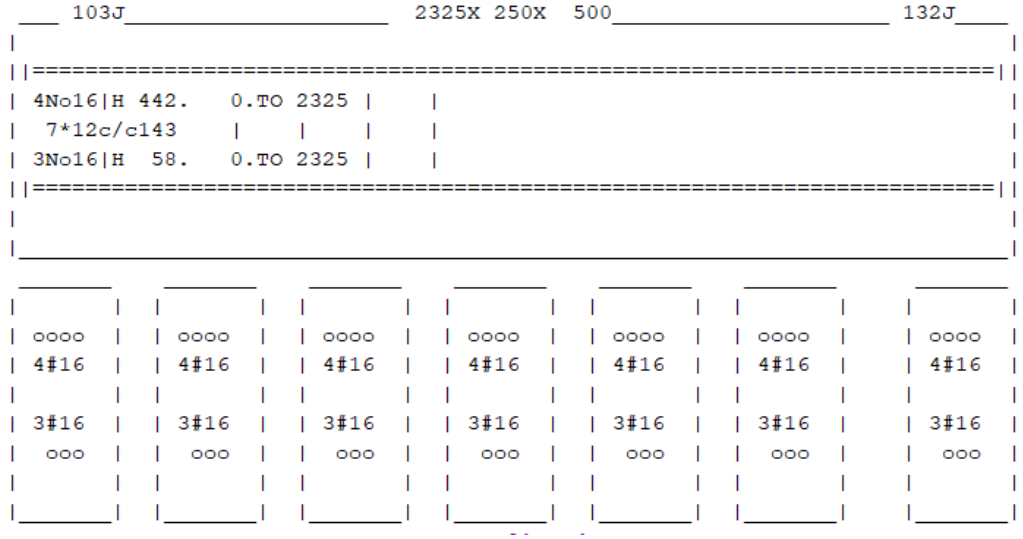
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9.8.1. Staad Output For B2

```
  AT END   SUPPORT - Vu=   68.14 KNS  Tu=   23.2 KN-MET
                     Vc=  103.4 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD    25 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
  AT START SUPPORT - Vu=   71.41 KNS  Vc=  101.04 KNS  Vs=    0.00 KNS
  Tu=   21.97 KN-MET  Tc=    3.6 KN-MET  Ts=   29.3 KN-MET  LOAD    27
  STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR  723. MM
  ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   7.24 SQ.CM.
```

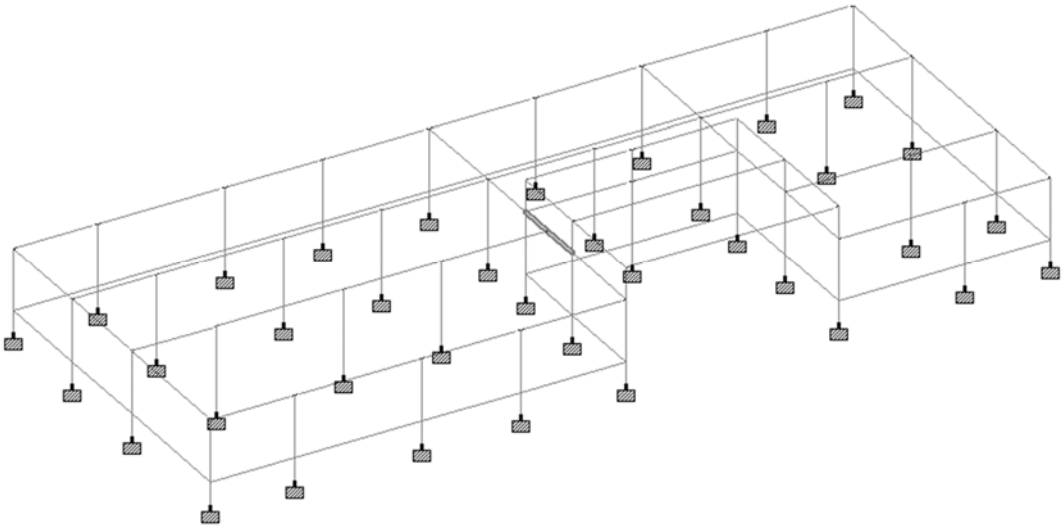


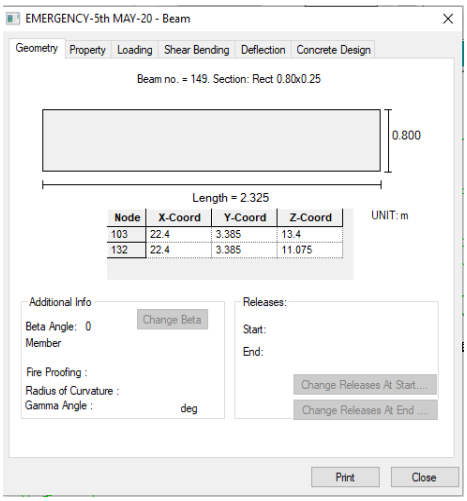
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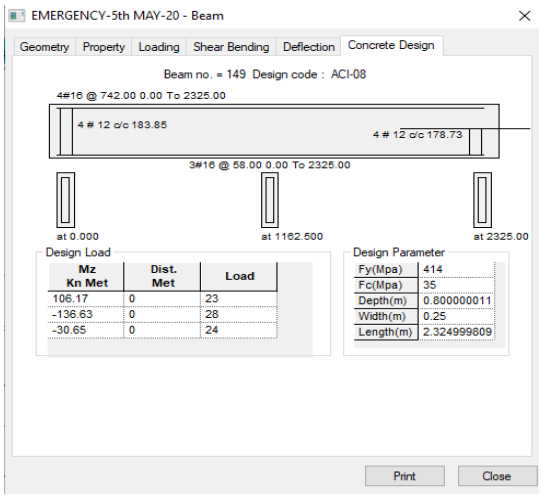
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9.9. Design of Beam-(B3)







Provide Equivalent 3-16 dia-Full Straight Bars & 3-16 Dia-Extra Reinf Bottom

& Equivalent 3-16 dia-Full Straight Bars & 2-16 Dia-Extra Reinf Top

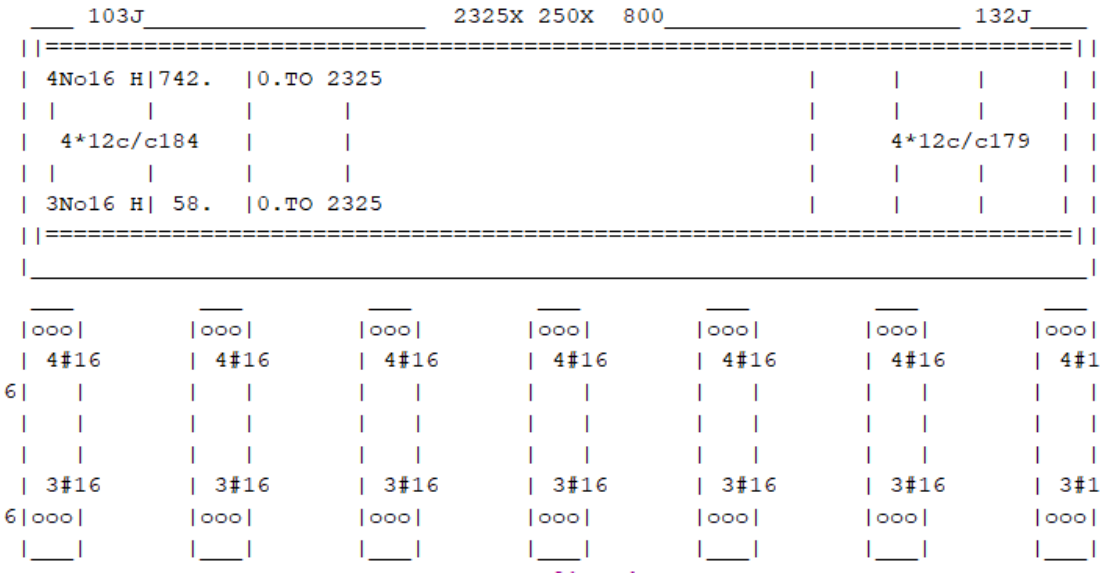
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9.9.1. Staad Output For B3

```
AT END   SUPPORT - Vu=   77.56 KNS  Vc=  174.92 KNS  Vs=    0.00 KNS
Tu=   37.91 KN-MET  Tc=    6.8 KN-MET  Ts=   50.6 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 179. MM  C/C FOR  423. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =  11.01 SQ.CM.
```



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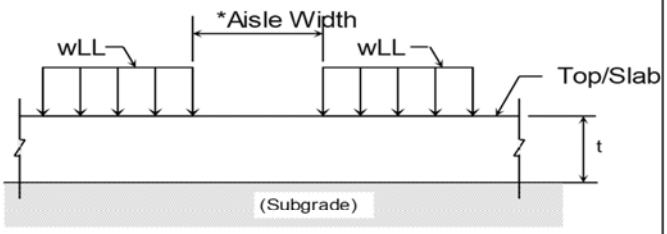
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

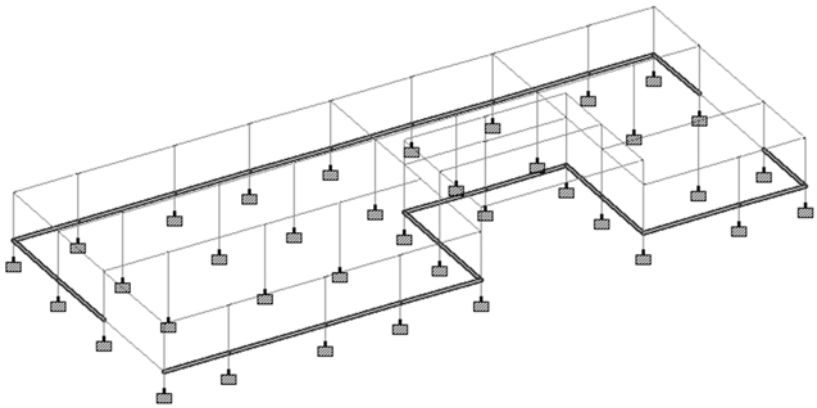
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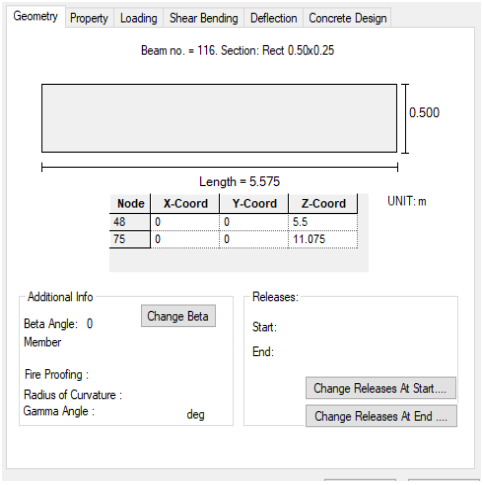
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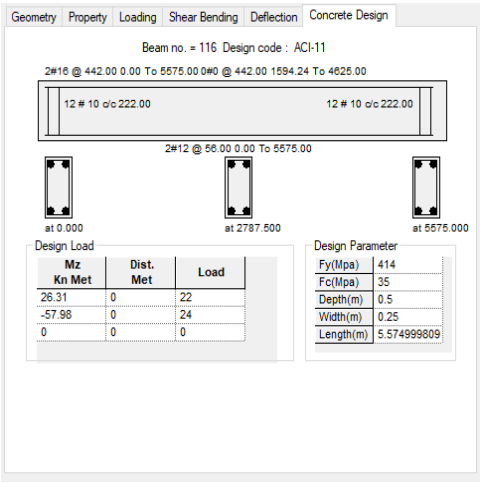
11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)



MOMENT ENVELOPE FOR BEAMS GB1





Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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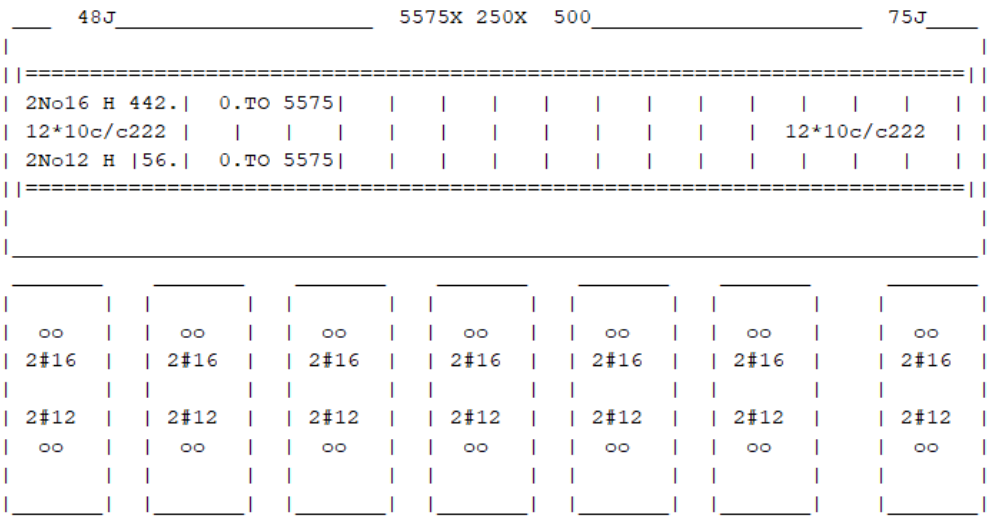
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11.1.1. STAAD OUTPUT FOR GB1

```
                B E A M  N O.   116 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   48.29 KNS  Vc=  106.75 KNS  Vs=    0.00 KNS
 Tu=    0.04 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD    22
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 2348. MM

 AT END   SUPPORT - Vu=   49.99 KNS  Vc=  106.41 KNS  Vs=    0.00 KNS
 Tu=    0.04 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD    22
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 2348. MM
```

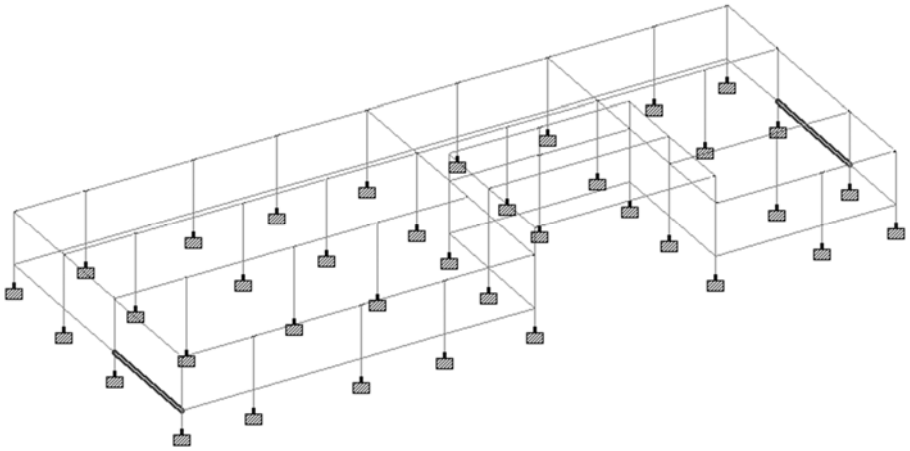


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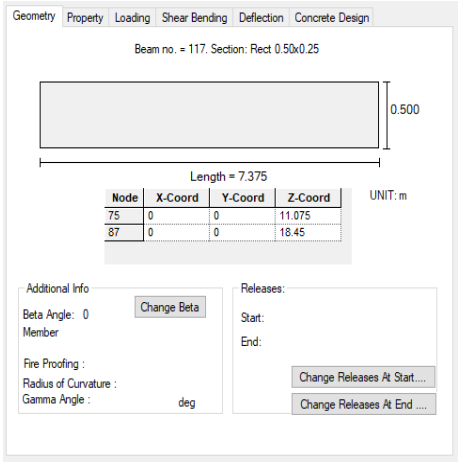
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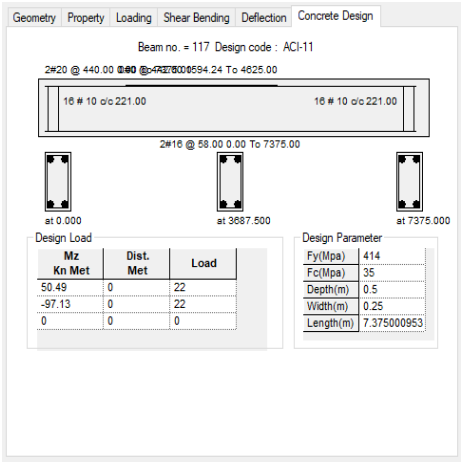
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11.2. Grade Beam(GB2)



MOMENT ENVELOPE FOR BEAMS GB2





Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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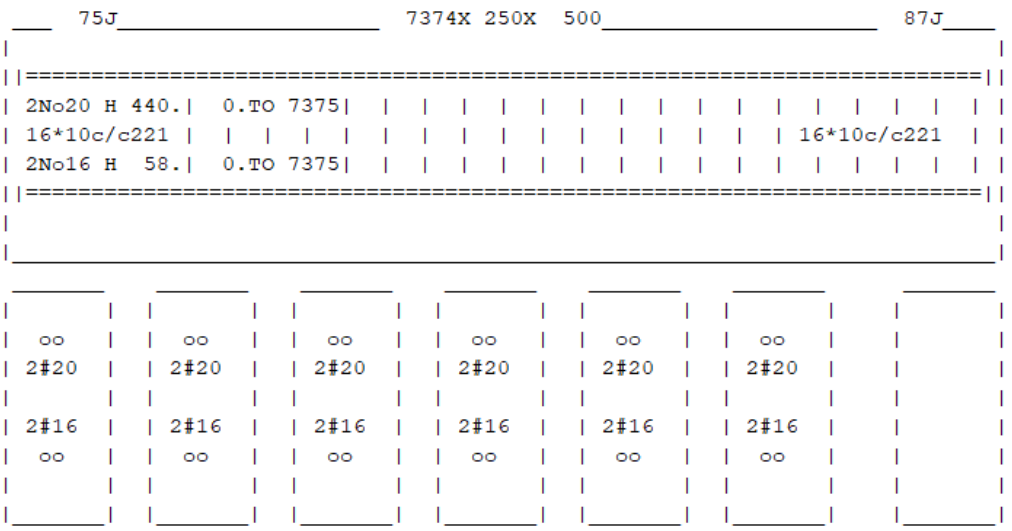
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11.2.1. STAAD OUTPUT FOR GB2

```
B E A M  N O.   117 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   49.58 KNS  Vc=  106.92 KNS  Vs=    0.00 KNS
Tu=    0.24 KN-MET  Tc=    4.1 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 3247. MM

AT END   SUPPORT - Vu=   47.53 KNS  Vc=  107.50 KNS  Vs=    0.00 KNS
Tu=    0.24 KN-MET  Tc=    4.1 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 3247. MM
```

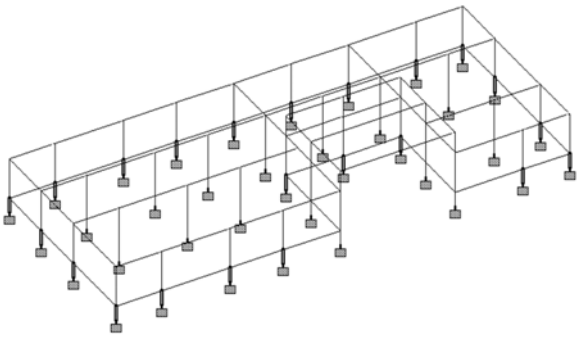


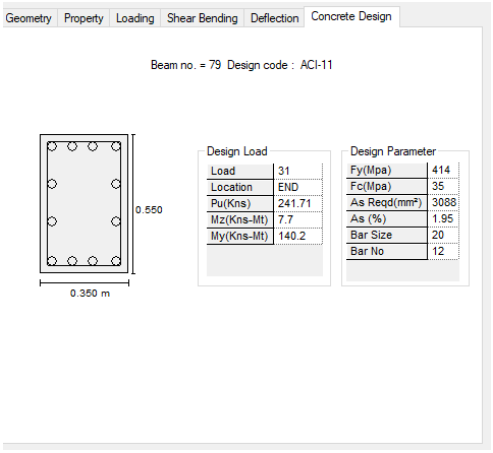
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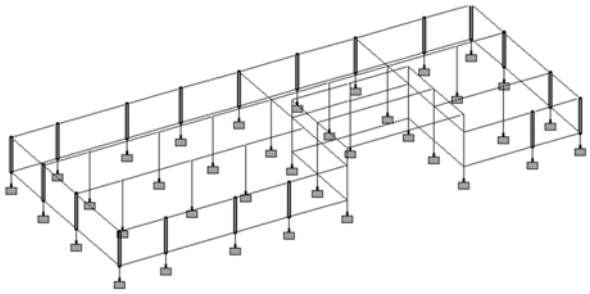
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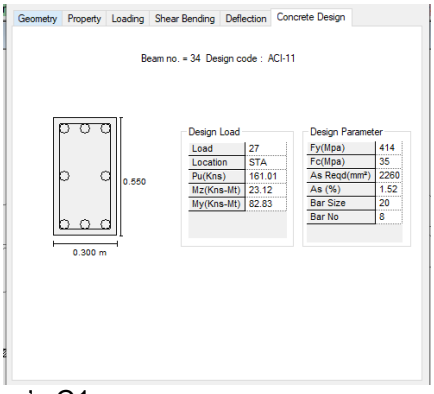
12. CONCRETE COLUMN DESIGN





Column's CR



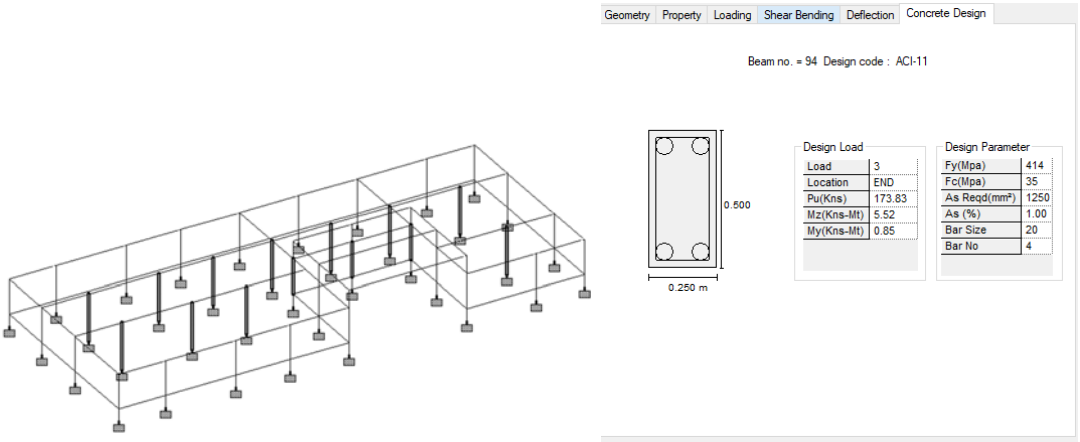


Column's C1

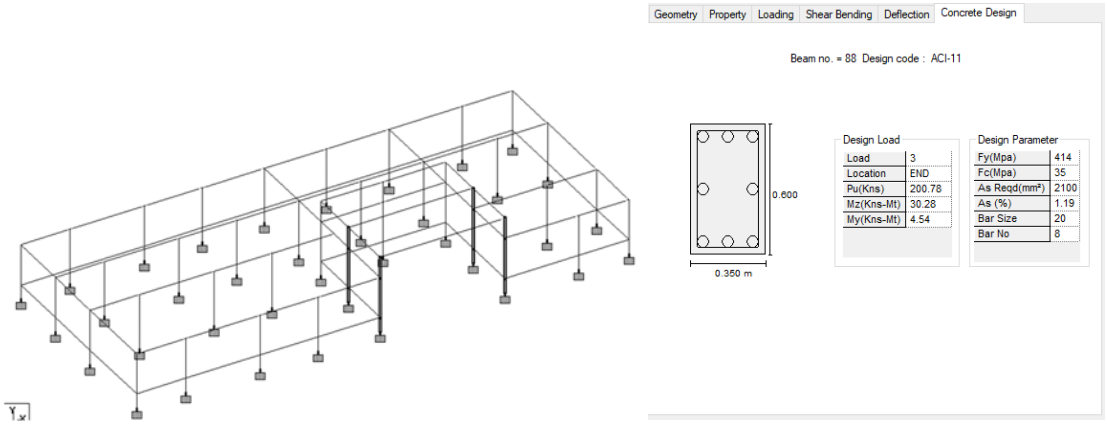
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Column's C2



Column's C3

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12.1. Staad Output For Column CR & C1

```
      ACI 318-11   COLUMN  NO.    79   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 350.0 X 550.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3087.7  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

 12 - 20 MM               1.958       31      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

      ACI 318-11   COLUMN  NO.    34   DESIGN RESULTS
      ================================================


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 300.0 X 550.0 MMS, TIED
        AREA OF STEEL REQUIRED =  2260.5  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 -------------------------------------------------------------

   8 - 20 MM              1.523       27      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 300.00 MM
```


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12.2. Staad Output For Column C2

```
      ACI 318-11   COLUMN  NO.    94   DESIGN RESULTS
      ==============================================


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
         ONLY MINIMUM STEEL IS REQUIRED.
         AREA OF STEEL REQUIRED =  1250.0  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

   4 - 20 MM               1.005        3      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 250.00 MM
```

12.3. Staad Output For Column C3

```
      ACI 318-11   COLUMN  NO.    88   DESIGN RESULTS
      ==============================================


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 350.0 X 600.0 MMS, TIED
         ONLY MINIMUM STEEL IS REQUIRED.
         AREA OF STEEL REQUIRED =  2100.0  SQ. MM
-------------------------------< PAGE 27 Ends Here >----------------------------
  STAAD SPACE                                                -- PAGE NO.   28


 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

   8 - 20 MM               1.197        3      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 320.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		
Calculations					
Shear force Vu = U*le	**8.096**	**kN**			
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**			

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage
$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000
Where:-
ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318
$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.) BUILDING PLANS	MD-216-4000-EG-AR-GAD-3043
2.) BUILDING SECTIONS & ELEVATIONS	MD-216-4000-EG-AR-GAD-3045
3.) LAYOUT OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-4000-EG-CV-GAD-1081
4.) DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-4000-EG-CV-DED-1082
5.) LAYOUT OF ROOF & GRADE SLAB	MD-216-4000-EG-CV-DED-1083
6.) GEOTECHNICAL REPORT	MD-216-3000-EG-CV-CAL-1008
7.) STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 4 0 3.385 0; 23 0 -1.2 0; 24 4.55 0 0; 25 4.55 3.385 0;
26 4.55 -1.2 0; 27 11.4 0 0; 28 11.4 3.385 0; 29 11.4 -1.2 0; 30 16.7 0 0;
31 16.7 3.385 0; 32 16.7 -1.2 0; 33 22.4 0 0; 34 22.4 3.385 0; 35 22.4 -1.2 0;
36 28.125 0 0; 37 28.125 3.385 0; 38 28.125 -1.2 0; 39 33.85 0 0;
40 33.85 3.385 0; 41 33.85 -1.2 0; 42 40.625 0 0; 43 40.625 3.385 0;
44 40.625 -1.2 0; 45 45.25 0 0; 46 45.25 3.385 0; 47 45.25 -1.2 0; 48 0 0 5.5;
49 0 3.385 5.5; 50 0 -1.2 5.5; 52 4.55 3.385 5.5; 53 4.55 -1.2 5.5;
55 11.4 3.385 5.5; 56 11.4 -1.2 5.5; 58 16.7 3.385 5.5; 59 16.7 -1.2 5.5;
61 22.4 3.385 5.5; 62 22.4 -1.2 5.5; 64 28.125 3.385 5.5; 65 28.125 -1.2 5.5;
67 33.85 3.385 5.5; 68 33.85 -1.2 5.5; 70 40.625 3.385 5.5; 71 40.625 -1.2 5.5;
72 45.25 0 5.5; 73 45.25 3.385 5.5; 74 45.25 -1.2 5.5; 75 0 0 11.075;
76 0 3.385 11.075; 77 0 -1.2 11.075; 79 4.55 3.385 11.075; 80 4.55 -1.2 11.075;
82 11.4 3.385 11.075; 83 11.4 -1.2 11.075; 85 16.7 3.385 11.075;
86 16.7 -1.2 11.075; 87 0 0 18.45; 88 0 3.385 18.45; 89 0 -1.2 18.45;
90 4.55 0 18.45; 91 4.55 3.385 18.45; 92 4.55 -1.2 18.45; 93 11.4 0 18.45;
94 11.4 3.385 18.45; 95 11.4 -1.2 18.45; 96 16.7 0 18.45; 97 16.7 3.385 18.45;
98 16.7 -1.2 18.45; 99 22.4 0 18.45; 100 22.4 3.385 18.45; 101 22.4 -1.2 18.45;
102 22.4 0 13.4; 103 22.4 3.385 13.4; 104 22.4 -1.2 13.4; 105 22.4 0 9;
106 22.4 3.385 9; 107 22.4 -1.2 9; 108 33.85 0 18.45; 109 33.85 3.385 18.45;
110 33.85 -1.2 18.45; 111 33.85 0 13.4; 112 33.85 3.385 13.4;
113 33.85 -1.2 13.4; 114 33.85 0 9; 115 33.85 3.385 9; 116 33.85 -1.2 9;
117 28.125 0 9; 118 28.125 3.385 9; 119 28.125 -1.2 9; 120 40.625 0 18.45;
121 40.625 3.385 18.45; 122 40.625 -1.2 18.45; 124 40.625 3.385 13.4;
125 40.625 -1.2 13.4; 126 45.25 0 18.45; 127 45.25 3.385 18.45;
128 45.25 -1.2 18.45; 129 45.25 0 13.4; 130 45.25 3.385 13.4;
131 45.25 -1.2 13.4; 132 22.4 3.385 11.075; 133 22.4 5.1 18.45;
134 22.4 5.1 13.4; 135 22.4 5.1 9; 136 33.85 5.1 18.45; 137 33.85 5.1 13.4;
138 33.85 5.1 9; 139 28.125 5.1 9;
MEMBER INCIDENCES
3 1 4; 31 1 23; 32 24 25; 33 24 26; 34 27 28; 35 27 29; 36 30 31; 37 30 32;
38 33 34; 39 33 35; 40 36 37; 41 36 38; 42 39 40; 43 39 41; 44 42 43; 45 42 44;
46 45 46; 47 45 47; 48 48 49; 49 48 50; 50 52 53; 52 55 56; 54 58 59; 56 61 62;
58 64 65; 60 67 68; 62 70 71; 64 72 73; 65 72 74; 66 75 76; 67 75 77; 68 79 80;
70 82 83; 72 85 86; 74 87 88; 75 87 89; 76 90 91; 77 90 92; 78 93 94; 79 93 95;
80 96 97; 81 96 98; 82 99 100; 83 99 101; 84 102 103; 85 102 104; 86 105 106;
87 105 107; 88 108 109; 89 108 110; 90 111 112; 91 111 113; 92 114 115;
93 114 116; 94 117 118; 95 117 119; 96 120 121; 97 120 122; 98 124 125;
100 126 127; 101 126 128; 102 129 130; 103 129 131; 104 1 24; 105 24 27;
106 27 30; 107 30 33; 108 33 36; 109 36 39; 110 39 42; 111 42 45; 112 45 72;
113 72 129; 114 129 126; 115 1 48; 116 48 75; 117 75 87; 118 87 90; 119 90 93;
120 93 96; 121 96 99; 122 102 99; 123 102 105; 124 105 117; 125 117 114;
126 114 111; 127 111 108; 128 108 120; 129 120 126; 130 4 25; 131 25 28;
132 28 31; 133 31 34; 134 34 37; 135 37 40; 136 40 43; 137 43 46; 138 46 73;
139 73 130; 140 130 127; 141 4 49; 142 49 76; 143 76 88; 144 88 91; 145 91 94;
146 94 97; 147 97 100; 148 103 100; 149 103 132; 150 106 118; 151 118 115;
152 115 112; 153 112 109; 154 109 121; 155 121 127; 156 49 52; 157 52 55;
158 55 58; 159 58 61; 160 61 64; 161 64 67; 162 67 70; 163 70 73; 164 76 79;
165 79 82; 166 82 85; 167 132 106; 168 85 132; 169 112 124; 170 124 130;
171 34 61; 172 106 61; 173 40 67; 174 115 67; 175 100 133; 176 103 134;
177 106 135; 178 109 136; 179 112 137; 180 115 138; 181 118 139; 183 134 133;
184 134 135; 185 135 139; 186 139 138; 187 138 137; 188 137 136; 189 134 137;
190 133 136;

```
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
48 TO 50 52 54 56 58 60 62 64 TO 68 70 72 86 87 92 TO 95 98 102 103 177 180 -
181 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
104 TO 129 PRIS YD 0.5 ZD 0.25
130 132 TO 135 137 138 140 TO 142 144 146 TO 148 150 TO 153 155 171 TO 174 -
183 TO 188 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
131 136 139 143 145 154 156 TO 166 168 TO 170 PRIS YD 0.55 ZD 0.25
189 PRIS YD 0.9 ZD 0.35
3 32 34 36 38 40 42 44 46 74 76 78 80 96 100 PRIS YD 0.55 ZD 0.3
82 TO 85 88 TO 91 175 176 178 179 PRIS YD 0.6 ZD 0.35
31 33 35 37 39 41 43 45 47 75 77 79 81 97 101 PRIS YD 0.55 ZD 0.35
190 PRIS YD 0.8 ZD 0.35
MEMBER PROPERTY
149 167 PRIS YD 0.8 ZD 0.25 CONSTANTS
BETA 90 MEMB 3 31 46 TO 50 52 54 56 60 64 TO 67 70 72 84 TO 87 90 TO 93 98 -
176 177 179 180
MATERIAL CONCRETE ALL
SUPPORTS
23 26 29 32 35 38 41 44 47 50 53 56 59 62 65 68 71 74 77 80 83 86 89 92 95 -
98 101 104 107 110 113 116 119 122 125 128 131 FIXED
******************************************************************************
MEMBER CRACKED
104 TO 174 183 TO 190 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
3 31 TO 50 52 54 56 58 60 62 64 TO 68 70 72 74 TO 98 100 TO 103 175 TO 180 -
181 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
MEMBER OFFSET
130 TO 137 144 TO 147 151 154 155 186 START 0.225 0 0
130 TO 137 144 TO 147 150 154 155 185 END -0.225 0 0
150 156 TO 166 168 TO 170 185 START 0.125 0 0
151 156 TO 166 168 TO 170 186 END -0.125 0 0
138 TO 143 START 0 0 0.225
138 TO 143 END 0 0 -0.225
190 START 0.3 0 0
190 END -0.3 0 0
189 START 0.175 0 0
189 END -0.175 0 0
DEFINE IBC 2003
SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.500 M 47.000 M 20.000 M 2.000 0.010 0 -
0 0 0 0 1.064 1.000 1.000 0.850 0 -
0 0 0 0.879 0.800 -0.550
```

```
!> END GENERATED DATA BLOCK
INT 1.65034 1.65034 1.65281 1.65524 1.65764 1.66001 1.66236 1.66467 1.66696 -
 1.66922 1.67146 1.67367 1.67586 1.67802 1.68016 HEIG  0  4.572  4.64338 -
 4.71477  4.78615  4.85754  4.92892  5.00031  5.07169  5.14308  5.21446 -
 5.28585  5.35723  5.42862  5.5
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.500 M 47.000 M 20.000 M 2.000 0.010 1 -
0 0 0 0 1.064 1.000 1.000 0.850 0 -
0 0 0 0.879 -0.282 0.550
!> END GENERATED DATA BLOCK
INT -1.07033 -1.07033 HEIG  0  5.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.500 M 47.000 M 20.000 M 2.000 0.010 2 -
0 0 0 0 1.064 1.000 1.000 0.850 0 -
0 0 0 0.879 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.56232 -1.56232 HEIG  0  5.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.500 M 20.000 M 47.000 M 2.000 0.010 0 -
0 0 0 0 1.064 1.000 1.000 0.850 0 -
0 0 0 0.857 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.62739 1.62739 1.62979 1.63216 1.6345 1.63682 1.6391 1.64136 1.64359 -
 1.6458 1.64798 1.65013 1.65226 1.65437 1.65646 HEIG  0  4.572  4.64338 -
 4.71477  4.78615  4.85754  4.92892  5.00031  5.07169  5.14308  5.21446 -
 5.28585  5.35723  5.42862  5.5
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.500 M 20.000 M 47.000 M 2.000 0.010 1 -
0 0 0 0 1.064 1.000 1.000 0.850 0 -
0 0 0 0.857 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.31182 -1.31182 HEIG  0  5.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.500 M 20.000 M 47.000 M 2.000 0.010 2 -
0 0 0 0 1.064 1.000 1.000 0.850 0 -
0 0 0 0.857 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.54158 -1.54158 HEIG  0  5.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
```

130 TO 137 UNI GY -17.6
144 TO 147 UNI GY -24.82
154 155 UNI GY -16.1
150 151 UNI GY -10.18
164 TO 166 168 UNI GY -43.57
156 TO 159 UNI GY -35.44
160 161 UNI GY -27.8
162 163 UNI GY -44.18
169 170 UNI GY -42.7
185 186 UNI GY -13.1
190 UNI GY -14.7
189 UNI GY -27.8
141 TO 143 152 153 183 184 UMOM GZ 1
144 TO 147 150 151 154 155 190 UMOM GX 1
138 TO 140 148 149 167 187 188 UMOM GZ -1
130 TO 137 185 186 UMOM GX -1
104 TO 121 128 129 UNI GY -12
122 TO 127 UNI GY -16
LOAD 4 LOADTYPE Live TITLE LL
MEMBER LOAD
130 TO 137 UNI GY -1.375
144 TO 147 UNI GY -1.85
154 155 UNI GY -1.26
150 151 UNI GY -0.875
164 TO 166 168 UNI GY -3.25
156 TO 159 UNI GY -2.77
160 161 UNI GY -2.25
162 163 UNI GY -3.35
169 170 UNI GY -3.23
185 186 UNI GY -1.125
190 UNI GY -1.26
189 UNI GY -2.385
41 TO 50 52 54 56 58 60 62 64 TO 68 70 72 74 TO 98 100 TO 143 152 153 183 -
184 UMOM GZ 0.1
144 TO 147 150 151 154 155 190 UMOM GX 0.1
138 TO 140 148 149 167 187 188 UMOM GZ -0.1
130 TO 137 185 186 UMOM GX -0.1
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 XR 0 0.5 YR 0 5.5
WIND LOAD -X 1 TYPE 2 XR 22.4 22.5 YR 0 5.5
WIND LOAD -Z 1 TYPE 3 YR 0 5.5 ZR 9 9.1
WIND LOAD -Z -1 TYPE 3 YR 0 5.5 ZR 0 0.1
WIND LOAD X 1 TYPE 1 XR 33.85 33.9 YR 0 5.5
WIND LOAD -X 1 TYPE 2 XR 45.25 47 YR 0 5.5
WIND LOAD -Z 1 TYPE 3 YR 0 5.5 ZR 18.45 18.5
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 5.5
WIND LOAD -Z 1 TYPE 5 YR 0 5.5
WIND LOAD -X 1 TYPE 6 XR 22.4 22.5 YR 0 5.5
WIND LOAD -X -1 TYPE 6 XR 33.85 33.9 YR 0 5.5
WIND LOAD -X 1 TYPE 6 XR 45.25 47 YR 0 5.5
WIND LOAD -X -1 TYPE 6 XR 0 0.1 YR 0 5.5
LOAD 7 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
3 32 34 36 38 40 42 44 46 48 64 66 74 76 78 80 82 84 86 88 90 92 94 96 100 -
102 130 TO 181 183 TO 190 TEMP 25
LOAD 77 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 5.5 ZR 18.45 18.5
WIND LOAD -Z -1 TYPE 5 YR 0 5.5

WIND LOAD -X -1 TYPE 6 XR 22.4 22.5 YR 0 5.5
WIND LOAD -X 1 TYPE 6 XR 33.85 33.9 YR 0 5.5
WIND LOAD -X 1 TYPE 6 XR 45.25 47 YR 0 5.5
WIND LOAD -X -1 TYPE 6 XR 0 0.1 YR 0 5.5
WIND LOAD Z -1 TYPE 4 YR 0 5.5 ZR 9 9.5
LOAD COMB 8 SLS: (DL)
3 1.0
LOAD COMB 9 SLS: (DL+T)
3 1.0 7 1.0
LOAD COMB 10 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 11 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 12 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 13 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 14 SLS: (D+0.75WX+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 16 SLS: (0.6D+WX)
3 0.6 5 1.0
LOAD COMB 17 SLS: (0.6D+WZ)
3 0.6 6 1.0
LOAD COMB 18 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 19 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 20 SLS: (D+WX)
3 1.0 5 1.0
LOAD COMB 21 SLS: (D+WZ)
3 1.0 6 1.0
LOAD COMB 22 ULS: (1.4 DL)
3 1.4
LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 29 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 30 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 31 ULS: (0.90D+1.6WX)
3 0.9 5 1.6
LOAD COMB 32 ULS: (0.90D+1.6WZ)
3 0.9 6 1.6
LOAD COMB 33 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 34 ULS: (0.90D+EX)
3 0.9 2 1.0

```
LOAD COMB 35 SLS: (DL+W-Z)
3 1.0 77 1.0
LOAD COMB 36 SLS: 0.6DL+WZ-
3 0.6 77 1.0
LOAD COMB 37 ULS: (0.9DL+1.6WZ-VE)
3 0.9 77 1.6
LOAD COMB 38 ULS:(1.2D+1.6(WZ-)+0.5LR
77 1.6 3 1.2 4 0.5
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MINMAIN 20 MEMB 3 31 TO 50 52 54 56 58 60 62 64 TO 68 70 72 74 TO 98 -
100 TO 103 131 136 145 154 156 TO 159 161 TO 166 168 TO 170 175 TO 181 185 -
186 189 190
MAXMAIN 20 MEMB 3 31 TO 50 52 54 56 58 60 62 64 TO 68 70 72 74 TO 98 -
100 TO 103 175 TO 181
MINSEC 10 ALL
DESIGN COLUMN 3 31 TO 50 52 54 56 58 60 62 64 TO 68 70 72 74 TO 98 -
100 TO 103 175 TO 181
SFACE 0.5 MEMB 130 132 TO 135 137 144 146 147 155
SFACE 0.55 MEMB 131 136 145 154
SFACE 0.8 MEMB 190
SFACE 0.9 MEMB 189
EFACE 0.5 MEMB 130 132 TO 135 137 144 146 147 155
EFACE 0.55 MEMB 131 136 145 154
EFACE 0.8 MEMB 190
EFACE 0.9 MEMB 189
DESIGN BEAM 104 TO 174 183 TO 190
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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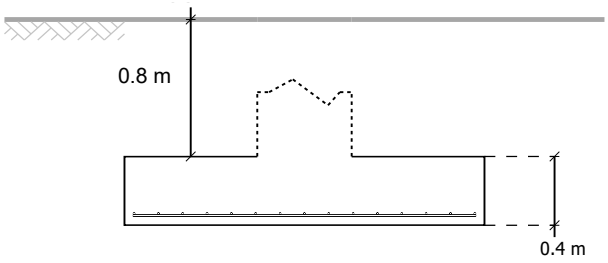
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Isolated Footing Design(ACI 318-08)

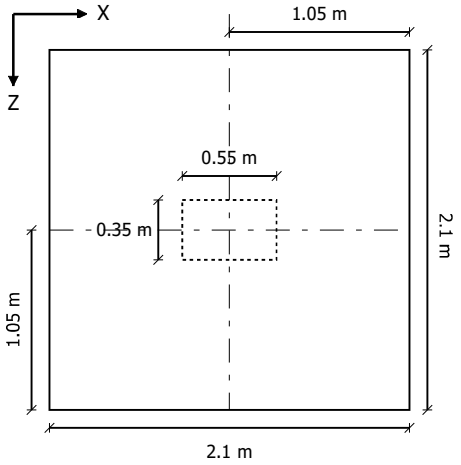
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
29	1	2.100m	2.100m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
29	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATIION-F1



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2100.000mm
Footing Width - Z (Fw) : 2100.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.100m

Initial Width (W_o) = 2.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00
38	ULS:(1.2D+1.6(WZ-)+0.5LR	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	236.721	10.460	0.053	-4.988	-4.370
9	236.047	7.884	0.071	-5.177	7.579
10	245.140	9.652	0.057	-5.495	-4.168
11	243.035	9.854	0.056	-5.369	-4.218
12	236.722	10.460	1.247	-0.635	-4.369
13	236.555	11.810	0.053	-4.992	-6.972
14	242.838	12.193	-23.854	-69.529	-7.988
15	243.026	9.085	24.929	61.788	-3.500
16	141.770	9.394	-31.848	-88.540	-7.649
17	142.020	5.250	33.195	86.549	-1.665
18	141.867	7.626	0.032	-2.996	-5.224
19	142.033	6.276	1.226	1.360	-2.621
20	236.459	13.578	-31.827	-90.535	-9.396
21	236.709	9.434	33.216	84.553	-3.412

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	331.410	14.644	0.075	-6.984	-6.117
23	334.674	10.634	0.101	-7.501	10.711
24	334.674	10.634	0.101	-7.501	10.711
25	297.325	13.754	-25.434	-75.235	-8.942
26	288.255	10.507	53.127	137.027	-3.611
27	287.855	17.137	-50.943	-143.114	-13.186
28	288.255	10.507	53.127	137.027	-3.611
29	283.828	14.481	0.063	-5.991	-8.961
30	284.066	12.552	1.769	0.233	-5.243
31	212.629	14.403	-50.961	-141.364	-11.976
32	213.029	7.773	53.109	138.777	-2.401
33	213.050	9.414	1.753	1.729	-3.932
34	212.812	11.343	0.047	-4.494	-7.650
35	236.703	9.385	-26.673	-77.250	-3.326
36	142.014	5.201	-26.694	-75.255	-1.578
37	213.020	7.694	-42.713	-120.108	-2.263
38	288.245	10.428	-42.695	-121.858	-3.473

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.410m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.822m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.100 m Governing Load Case : # 8

Width (W_2) = 2.100 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.410 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 42.335 kN

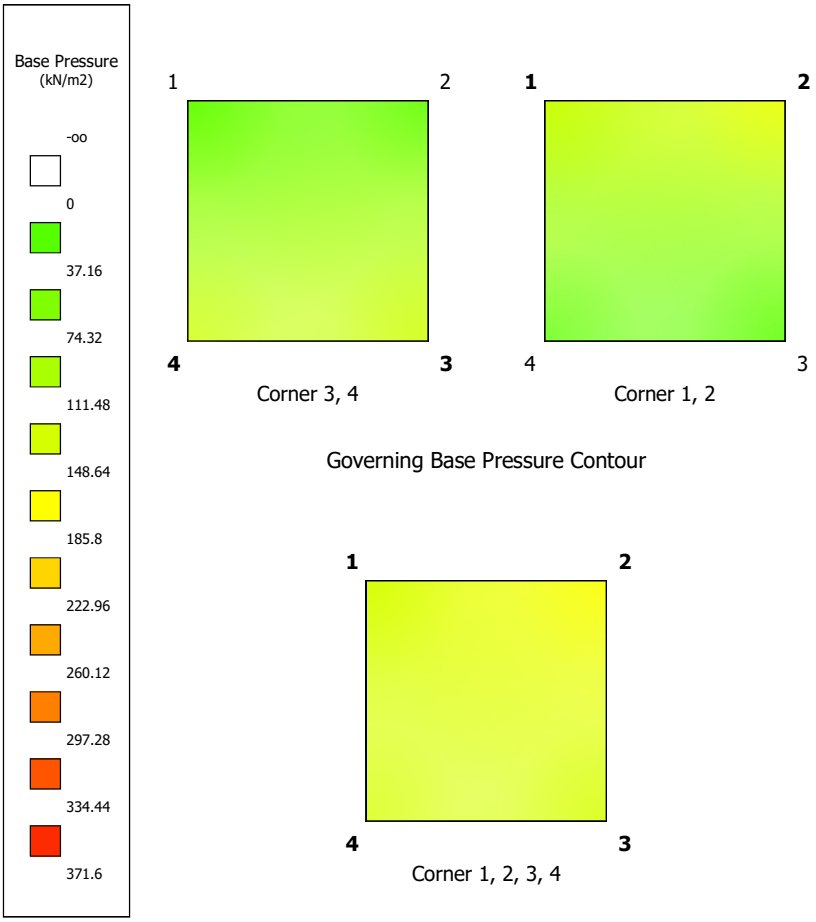
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 60.730 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



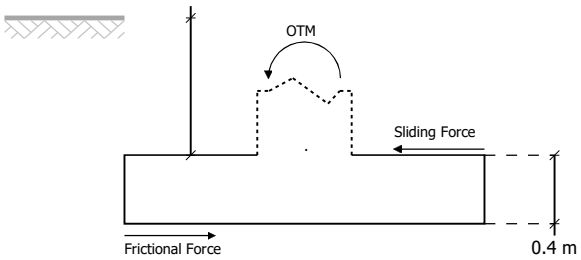
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
20	**134.2876**	153.5008	19.6912	0.4780	0.000
20	134.2876	**153.5008**	19.6912	0.4780	0.000
21	9.0009	18.3125	**145.0911**	135.7796	0.000
21	9.0009	18.3125	145.0911	**135.7796**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
20	**134.2876**	153.5008	19.6912	0.4780
20	134.2876	**153.5008**	19.6912	0.4780
21	9.0009	18.3125	**145.0911**	135.7796
21	9.0009	18.3125	145.0911	**135.7796**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	16.242	3190.714	22.970	1.500	72.226	41.783	1.500
9	21.506	2395.359	30.413	1.500	69.540	80.607	1.500
10	18.037	3033.137	25.508	1.500	67.171	45.615	1.500
11	17.561	3070.326	24.835	1.500	68.345	44.611	1.500
12	16.242	136.230	22.809	1.500	2631.204	41.785	1.500
13	14.378	3211.167	20.333	1.500	72.141	30.542	1.500
14	14.185	7.250	9.130	1.500	4.618	28.278	1.500
15	19.047	6.942	9.223	1.500	5.092	51.022	1.500
16	13.031	3.844	5.214	1.500	2.558	22.593	1.500
17	23.339	3.692	5.157	1.500	2.598	68.521	1.500
18	16.058	3877.576	22.709	1.500	86.858	31.155	1.500
19	19.527	99.974	27.103	1.500	140.142	50.270	1.500
20	12.503	5.334	6.938	1.500	3.471	24.085	1.500
21	18.007	5.115	6.958	1.500	3.667	49.732	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20
Governing Disturbing Force : 13.578kN
Governing Restoring Force : 169.762kN
Minimum Sliding Ratio for the Critical Load Case : 12.503
Critical Load Case for Overturning about X-Direction : 16
Governing Overturning Moment : -101.279kNm
Governing Resisting Moment : 259.056kNm
Minimum Overturning Ratio for the Critical Load Case : 2.558

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17
Governing Disturbing Force : 33.195kN
Governing Restoring Force : 122.542kN
Minimum Sliding Ratio for the Critical Load Case : 3.692
Critical Load Case for Overturning about Z-Direction : 16
Governing Overturning Moment : -11.406kNm
Governing Resisting Moment : 257.695kNm
Minimum Overturning Ratio for the Critical Load Case : 22.593

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 33.608kN

Governing Restoring Force : 173.301kN

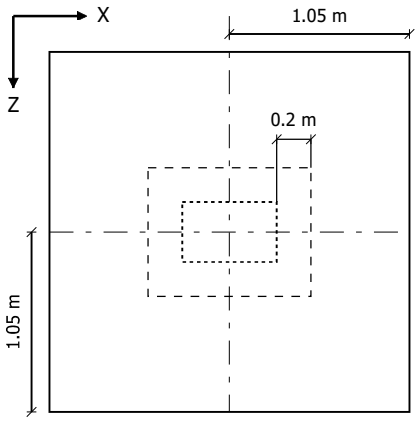
Minimum Sliding Ratio for the Critical Load Case : 5.157

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.571

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 376.154kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2439.551kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2146.805kN

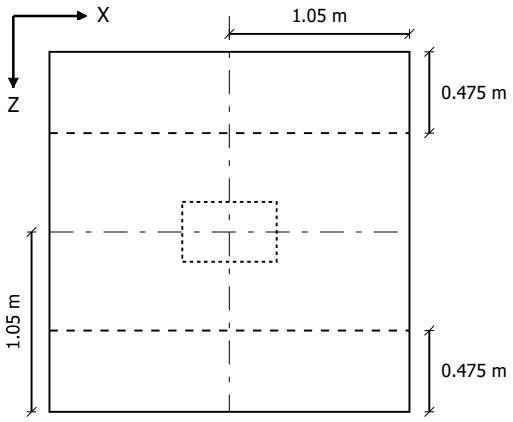
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.531m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

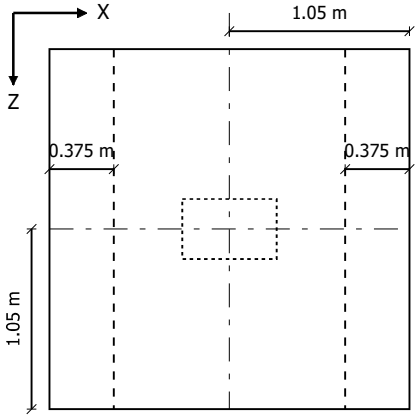
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{ux} is # 27 $V_{ux} = V_{ux}|_{x=D_x}$ = 187.096 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

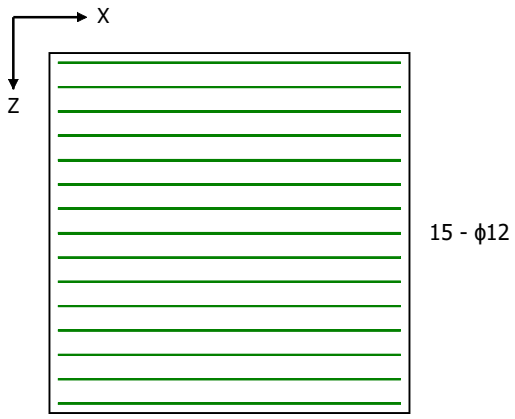
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{uz} is # 22 $V_{uz} = V_{uz}|_{z=D_z}$ = 94.752 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 22

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00159	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.325	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	65.825	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	73.139	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00070	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00061	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1333.030	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.725 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00235

From ACI Cl.7.6.1, minimum req'd clear distance between bars

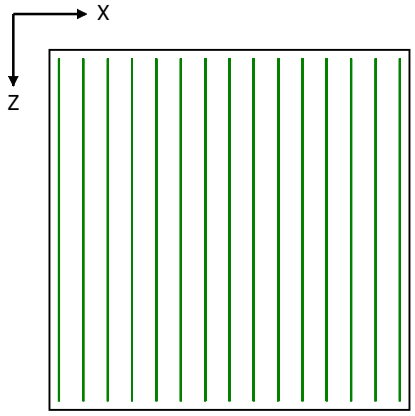
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.273

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00164

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.225 m

Ultimate moment, $M_u|_{x=D_x}$ = 133.409 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 148.232 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00154

(Based on gross depth) ρ x d_{eff} / Depth = 0.00128

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1381.116 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.825 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

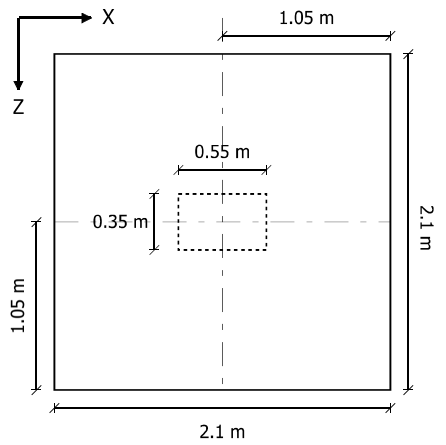
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00243

From ACI Cl.7.6.1, minimum req'd clear distance between bars

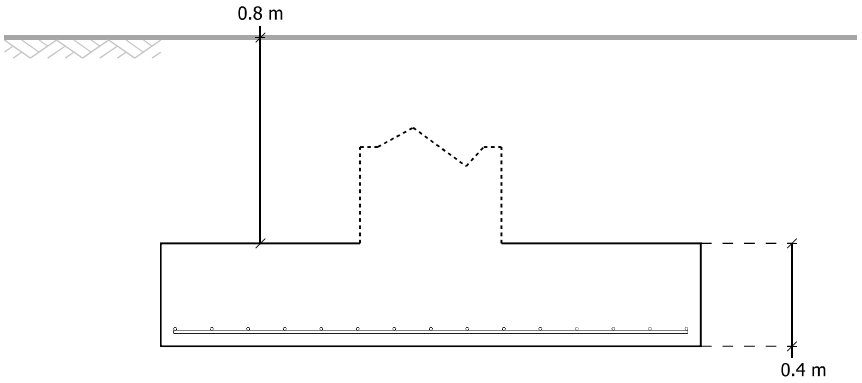
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.228

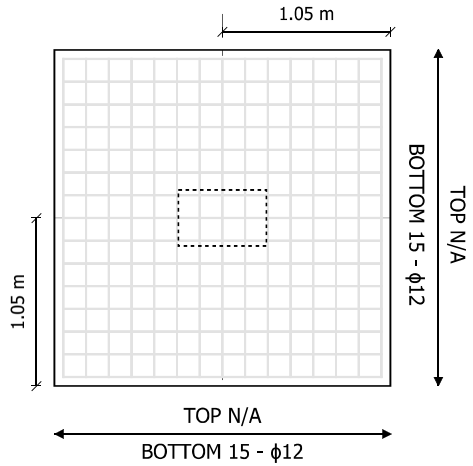
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.1 m	2.1 m	0.4 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

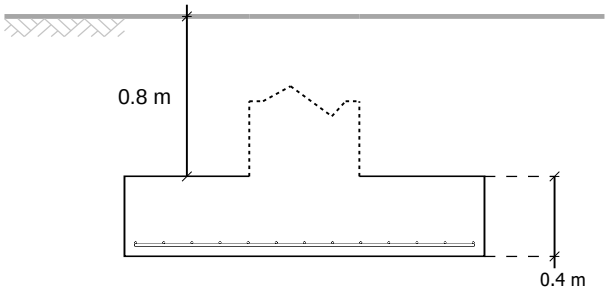
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

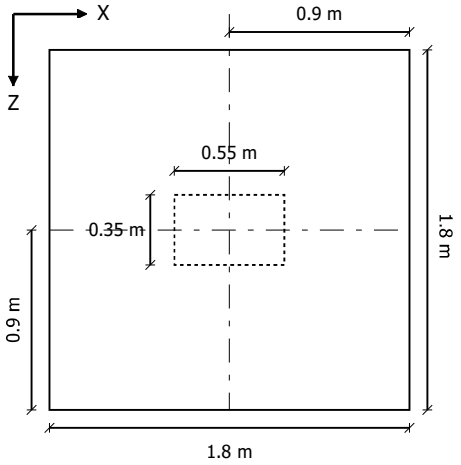
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
38	1	1.800m	1.800m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
38	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATIION-F2



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1800.000mm
Footing Width - Z (Fw) : 1800.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.800m

Initial Width (W_o) = 1.800m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00
38	ULS:(1.2D+1.6(WZ-)+0.5LR	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	220.943	-1.126	0.700	-1.099	0.999
9	219.370	0.923	0.187	-3.477	-5.342
10	228.754	-1.149	0.777	-1.210	0.992
11	226.801	-1.143	0.758	-1.182	0.994
12	220.940	-1.129	1.714	2.357	1.005
13	220.944	0.294	0.699	-1.101	-1.627
14	226.797	0.784	-16.595	-37.261	-2.370
15	226.762	-0.768	20.029	41.971	0.591
16	132.560	1.894	-22.717	-48.764	-3.885
17	132.513	-0.176	26.115	56.879	0.063
18	132.567	0.745	0.419	-0.662	-2.026
19	132.563	-0.679	1.434	2.796	0.605
20	220.937	1.443	-22.437	-49.204	-3.486
21	220.891	-0.626	26.395	56.439	0.462

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	309.320	-1.576	0.980	-1.538	1.399
23	311.023	1.280	0.300	-4.924	-7.483
24	311.023	1.280	0.300	-4.924	-7.483
25	277.624	0.668	-17.546	-39.980	-2.400
26	268.953	-0.563	41.991	90.687	0.337
27	269.028	2.749	-36.141	-78.342	-5.980
28	268.953	-0.563	41.991	90.687	0.337
29	265.133	0.678	0.839	-1.322	-2.552
30	265.128	-1.356	2.288	3.618	1.207
31	198.840	3.098	-36.389	-77.957	-6.276
32	198.765	-0.214	41.742	91.072	0.040
33	198.845	-1.018	2.078	3.948	0.908
34	198.850	1.016	0.629	-0.993	-2.852
35	220.915	-0.610	-20.256	-48.640	0.429
36	132.538	-0.159	-20.536	-48.201	0.030
37	198.805	-0.188	-32.900	-77.055	-0.013
38	268.993	-0.537	-32.651	-77.440	0.284

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.240m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.404m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.800 m Governing Load Case : # 8

Width (W_2) = 1.800 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 26

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.240 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 31.103 kN

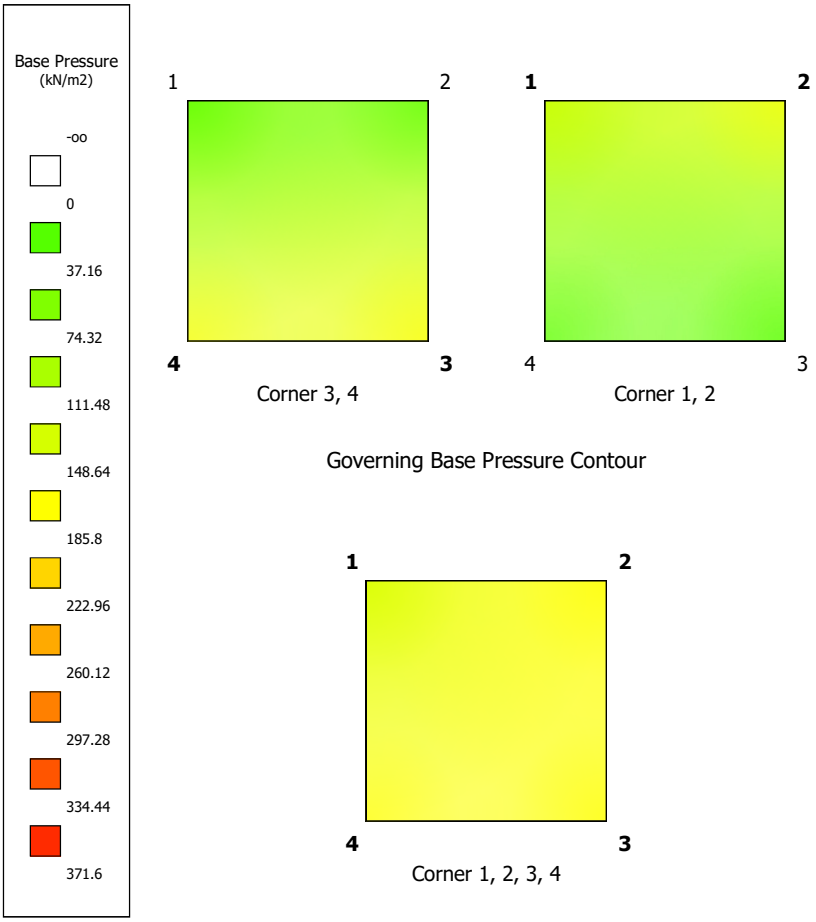
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 43.882 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



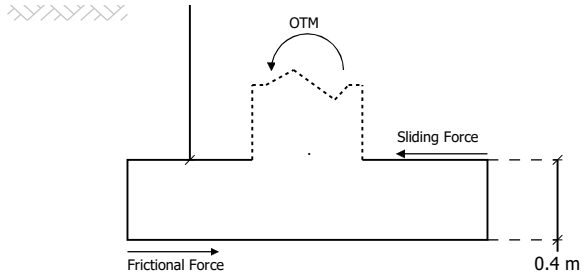
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
20	**147.0093**	155.3696	35.6591	27.2987	0.000
20	147.0093	**155.3696**	35.6591	27.2987	0.000
21	23.1249	21.6581	**159.5145**	160.9814	0.000
21	23.1249	21.6581	159.5145	**160.9814**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
20	**147.0093**	155.3696	35.6591	27.2987
20	147.0093	**155.3696**	35.6591	27.2987
21	23.1249	21.6581	**159.5145**	160.9814
21	23.1249	21.6581	159.5145	**160.9814**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	139.310	224.167	167.333	1.500	327.358	184.116	1.500
9	169.106	835.032	234.393	1.500	78.360	46.477	1.500
10	140.166	207.152	164.174	1.500	305.967	188.709	1.500
11	139.955	211.080	164.960	1.500	310.950	187.562	1.500
12	138.908	91.521	108.081	1.500	88.095	183.211	1.500
13	532.771	224.365	292.427	1.500	326.153	152.979	1.500
14	203.967	9.638	13.615	1.500	6.226	101.412	1.500
15	208.181	7.985	11.284	1.500	5.467	302.839	1.500
16	58.083	4.842	6.824	1.500	3.258	40.347	1.500
17	624.862	4.211	5.955	1.500	2.799	1406.990	1.500
18	147.708	262.421	182.035	1.500	381.343	80.598	1.500
19	162.062	76.718	98.062	1.500	55.932	213.643	1.500
20	108.652	6.990	9.865	1.500	4.607	65.680	1.500
21	250.366	5.941	8.400	1.500	4.000	374.281	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 1.894kN
Governing Restoring Force : 109.999kN
Minimum Sliding Ratio for the Critical Load Case : 58.083
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 67.325kNm
Governing Resisting Moment : 188.446kNm
Minimum Overturning Ratio for the Critical Load Case : 2.799

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17
Governing Disturbing Force : 26.115kN
Governing Restoring Force : 109.974kN
Minimum Sliding Ratio for the Critical Load Case : 4.211
Critical Load Case for Overturning about Z-Direction : 16
Governing Overturning Moment : -4.643kNm
Governing Resisting Moment : 187.322kNm
Minimum Overturning Ratio for the Critical Load Case : 40.347

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 26.116kN

Governing Restoring Force : 155.527kN

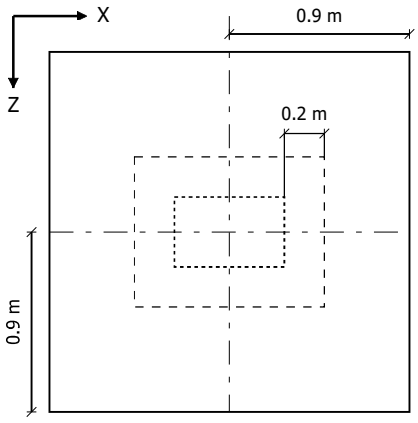
Minimum Sliding Ratio for the Critical Load Case : 5.955

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.571

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 312.091kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2439.551kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2146.805kN

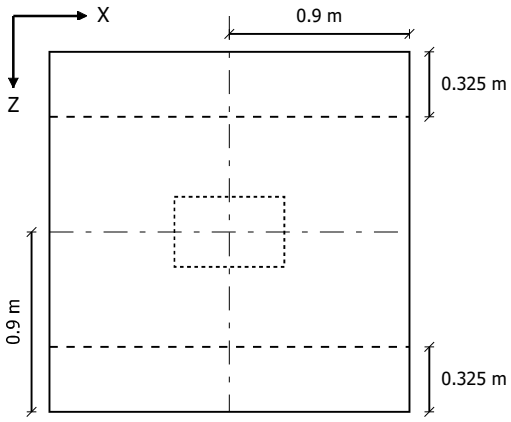
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

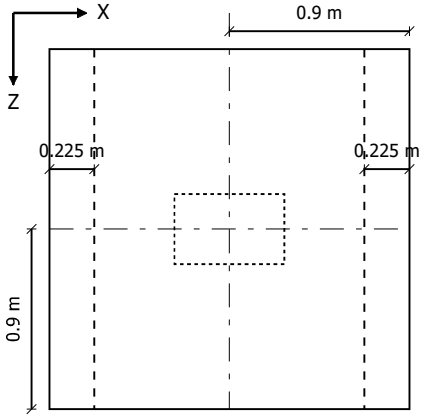
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 132.489 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.281 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

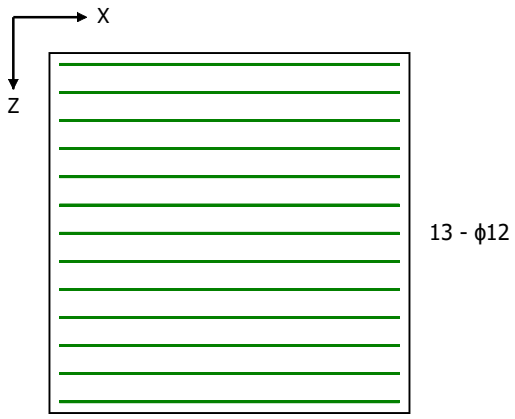
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 63.771 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00165
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.175 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	44.110 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	49.011 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00055
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00047
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1190.735 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}} =$ 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

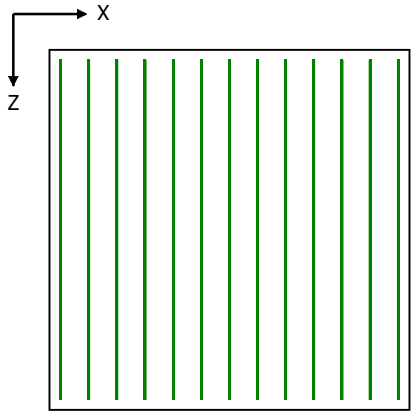
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.235

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00170

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.075 m

Ultimate moment, $M_u|_{x=D_x}$ = 88.663 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 98.514 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00119

(Based on gross depth) ρ x d_{eff} / Depth = 0.00099

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1223.068 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

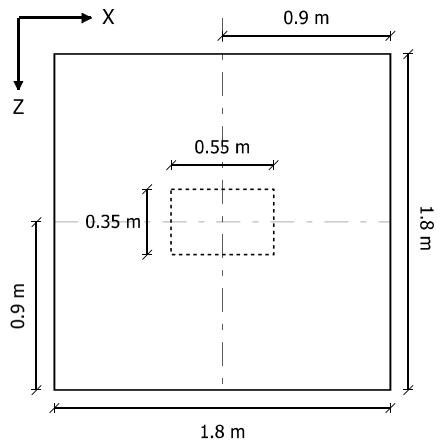
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00246

From ACI Cl.7.6.1, minimum req'd clear distance between bars

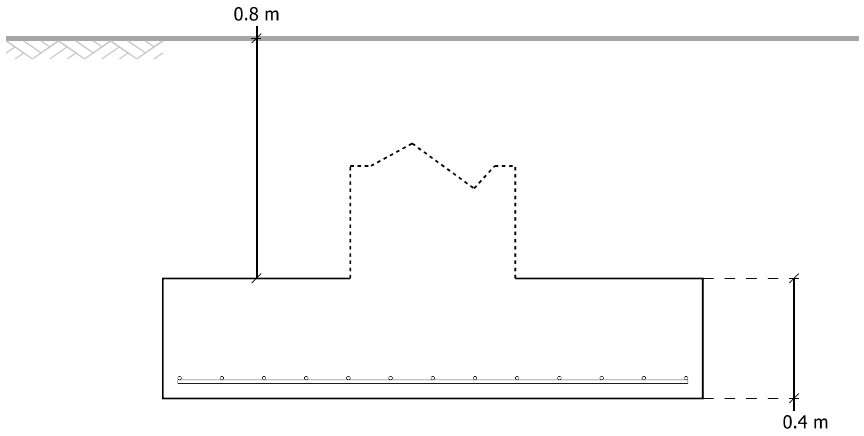
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.202

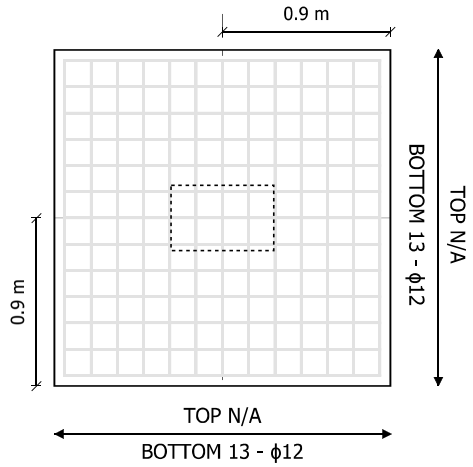
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.8 m	1.8 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

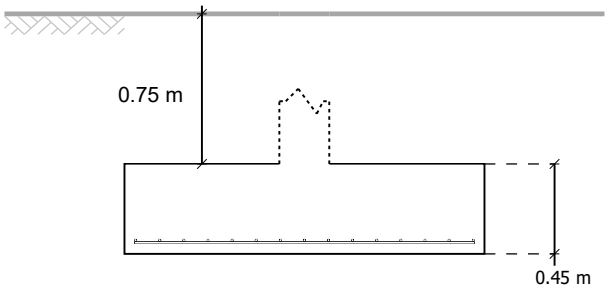
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

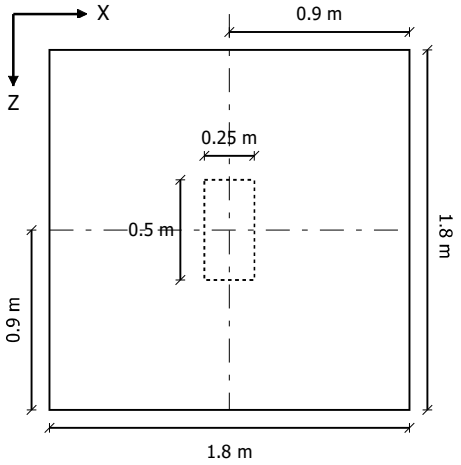
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
83	1	1.800m	1.800m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
83	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATIION-F3



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 1800.000mm
Footing Width - Z (Fw) : 1800.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 125.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.800m

Initial Width (W_o) = 1.800m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00
38	ULS:(1.2D+1.6(WZ-)+0.5LR	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	295.654	7.313	0.046	0.028	-11.526
9	295.208	6.489	0.041	0.040	-9.626
10	315.226	7.865	0.048	0.023	-12.198
11	310.333	7.727	0.048	0.024	-12.030
12	295.656	7.285	1.126	4.954	-11.461
13	295.773	8.276	0.046	0.042	-13.784
14	310.766	10.757	0.001	-0.111	-19.129
15	310.184	6.581	-0.062	-0.173	-9.357
16	177.969	8.428	-0.035	-0.164	-16.380
17	177.193	2.861	-0.118	-0.246	-3.352
18	177.512	5.351	0.028	0.031	-9.174
19	177.394	4.360	1.108	4.943	-6.850
20	296.231	11.353	-0.017	-0.152	-20.991
21	295.455	5.786	-0.100	-0.234	-7.962

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	413.916	10.238	0.065	0.039	-16.136
23	423.077	9.361	0.059	0.053	-13.812
24	423.077	9.361	0.059	0.053	-13.812
25	386.562	12.890	0.009	-0.119	-22.478
26	364.252	6.608	-0.177	-0.389	-8.465
27	365.494	15.515	-0.044	-0.258	-29.311
28	364.252	6.608	-0.177	-0.389	-8.465
29	354.955	10.151	0.056	0.054	-17.058
30	354.788	8.735	1.599	7.071	-13.738
31	267.012	13.046	-0.059	-0.264	-25.517
32	265.769	4.138	-0.192	-0.395	-4.671
33	266.091	6.541	1.585	7.063	-10.280
34	266.259	7.957	0.042	0.046	-13.600
35	295.488	5.949	0.197	0.377	-8.347
36	177.226	3.024	0.178	0.365	-3.737
37	265.822	4.400	0.282	0.583	-5.288
38	364.305	6.869	0.297	0.589	-9.082

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.240 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.260 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.800 m Governing Load Case : # 8

Width (W_2) = 1.800 m Governing Load Case : # 8

Depth (D_2) = 0.450 m Governing Load Case : # 23

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.240 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 34.991 kN

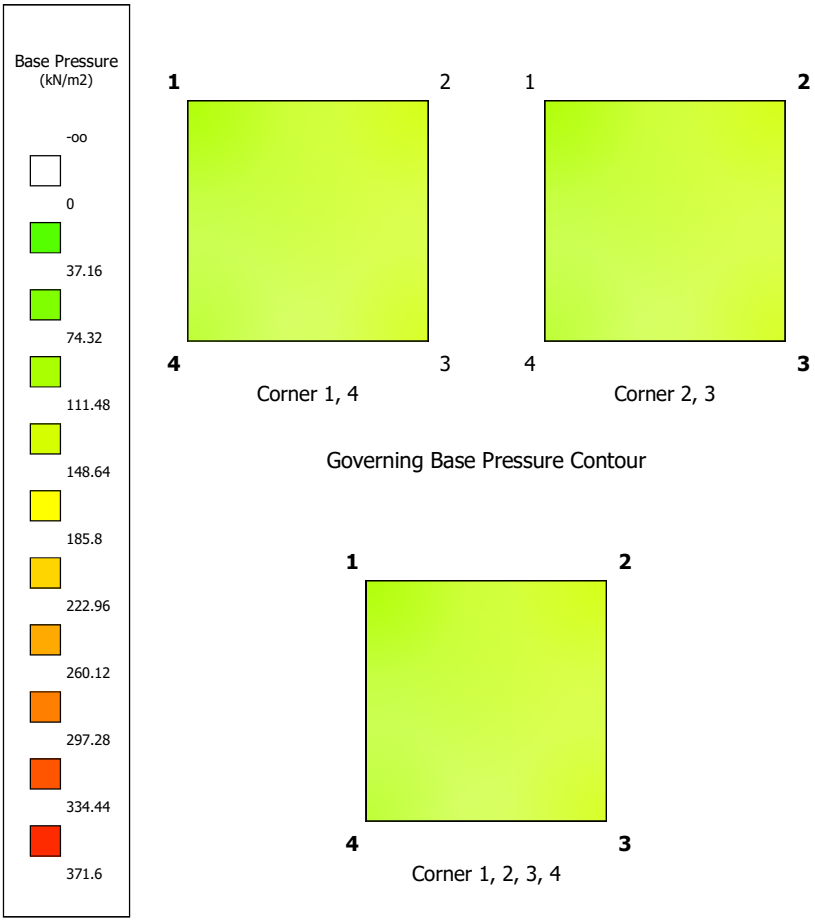
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 42.051 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



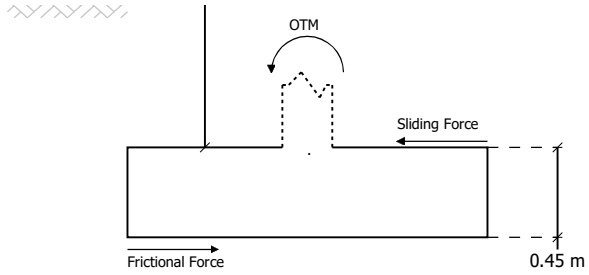
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
15	**107.0467**	132.3941	131.9813	106.6339	0.000
14	95.1482	**144.4679**	144.2392	94.9195	0.000
14	95.1482	144.4679	**144.2392**	94.9195	0.000
15	107.0467	132.3941	131.9813	**106.6339**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
15	**107.0467**	132.3941	131.9813	106.6339
14	95.1482	**144.4679**	144.2392	94.9195
14	95.1482	144.4679	**144.2392**	94.9195
15	107.0467	132.3941	131.9813	**106.6339**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	25.483	4033.160	36.038	1.500	6887.142	22.715	1.500
9	28.682	4536.091	40.562	1.500	5732.296	26.794	1.500
10	24.939	4048.426	35.268	1.500	7953.788	22.506	1.500
11	25.068	4044.765	35.450	1.500	7668.073	22.556	1.500
12	25.581	165.445	35.752	1.500	61.420	22.835	1.500
13	22.525	4008.937	31.854	1.500	5313.880	19.229	1.500
14	18.026	367242.051	25.493	1.500	3140.413	14.609	1.500
15	29.418	3142.655	41.602	1.500	1737.166	28.383	1.500
16	15.129	3599.614	21.396	1.500	1277.988	11.434	1.500
17	44.437	1074.806	62.790	1.500	765.544	49.569	1.500
18	23.787	4543.000	33.639	1.500	5247.229	19.879	1.500
19	29.181	114.832	39.997	1.500	42.081	26.115	1.500
20	16.440	11017.498	23.249	1.500	2098.285	12.915	1.500
21	32.191	1866.439	45.519	1.500	1199.856	31.838	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 8.428kN
Governing Restoring Force : 127.506kN
Minimum Sliding Ratio for the Critical Load Case : 15.129
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 5.442kNm
Governing Resisting Moment : 228.988kNm
Minimum Overturning Ratio for the Critical Load Case : 42.081

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19
Governing Disturbing Force : 1.108kN
Governing Restoring Force : 127.218kN
Minimum Sliding Ratio for the Critical Load Case : 114.832
Critical Load Case for Overturning about Z-Direction : 16
Governing Overturning Moment : -20.173kNm
Governing Resisting Moment : 230.645kNm
Minimum Overturning Ratio for the Critical Load Case : 11.434

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 16

Governing Disturbing Force : 8.428kN

Governing Restoring Force : 180.320kN

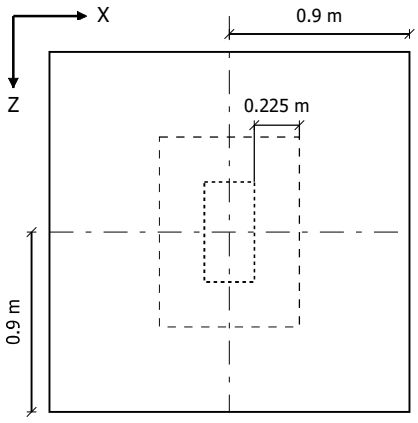
Minimum Sliding Ratio for the Critical Load Case : 21.396

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 411.250kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.076m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2381.421kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4241.035kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2381.421kN

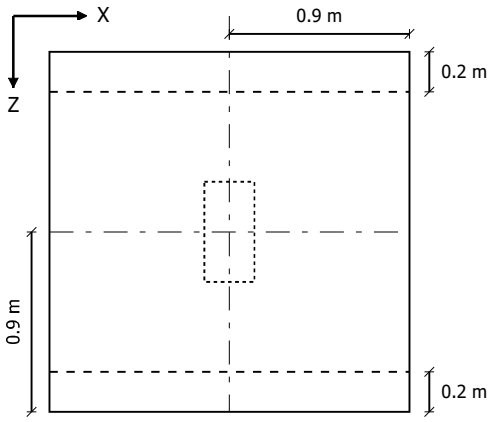
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1786.066kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 696.775kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.256m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

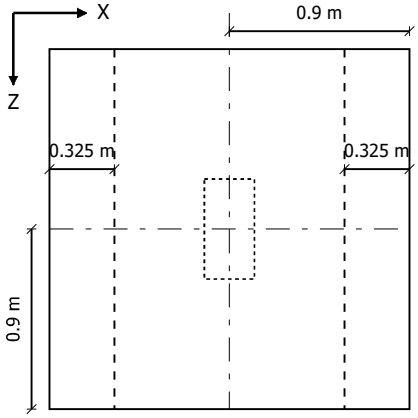
From above calculations, 0.75 X V_c = 522.581 kN

Critical load case for V_{ux} is # 23 $V_{ux} = V_{ux}|_{x=D_x}$ = 71.160 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 696.775 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.381 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

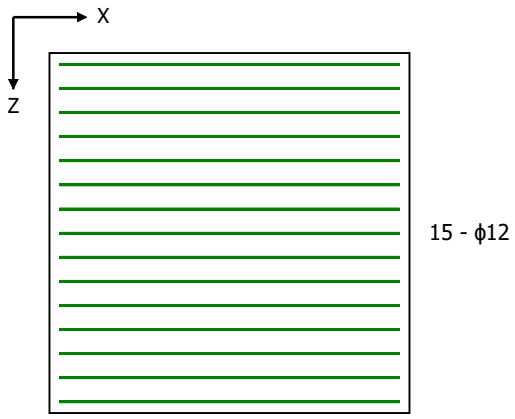
From above calculations, 0.75 X V_c = 522.581 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 115.884 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00171
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.025 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	90.591 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	100.656 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00086
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00076
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1384.891 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 125.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 125mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.725 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.394 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00239

From ACI Cl.7.6.1, minimum req'd clear distance between bars

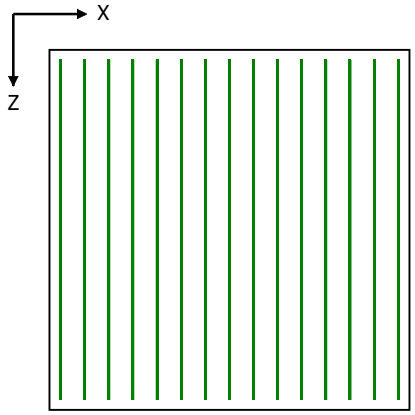
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.225

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.382 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00168

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.150	m
Ultimate moment,	$M_u\|_{x=D_x}$ =	58.717	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	65.241	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00059	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00050	
Since	$\rho = \rho_{min}$	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1360.188	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 125.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 125mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.600	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

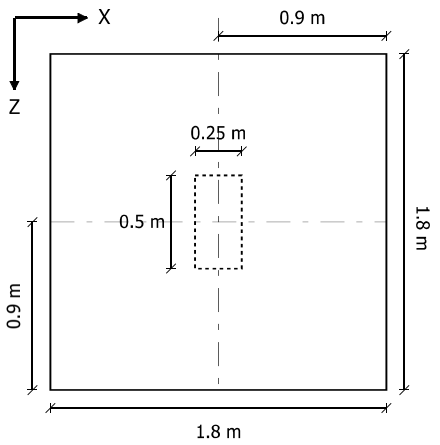
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.382	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00247	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

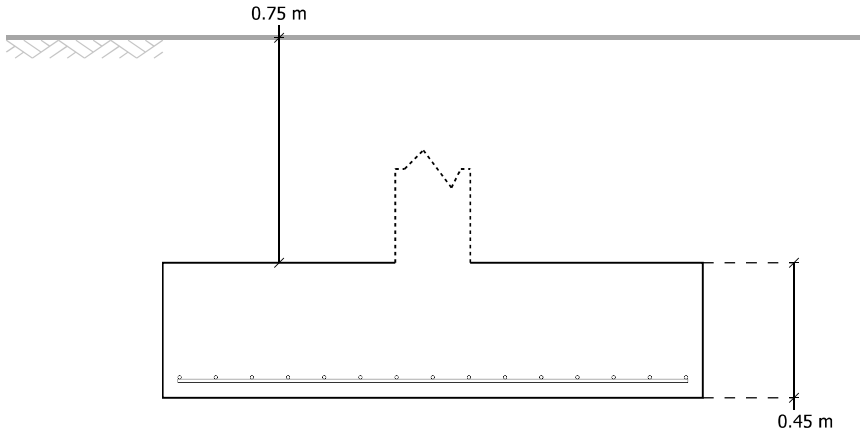
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.247

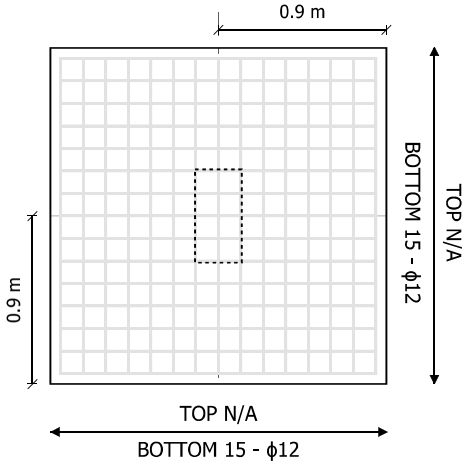
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.8 m	1.8 m	0.45 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

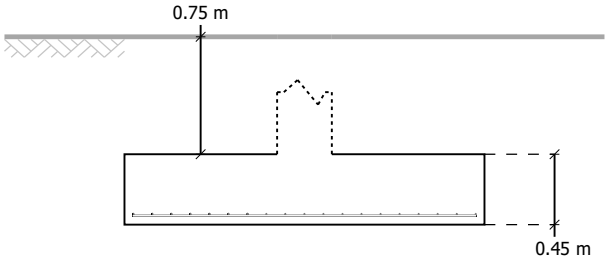
file:///C:/Staad%20Foundation%20Advanced%20SS3/CalcXsl/footing.xml

Isolated Footing Design(ACI 318-08)

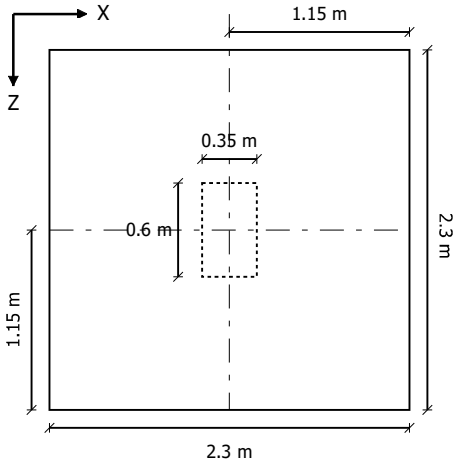
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
104	1	2.300m	2.300m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
104	19 - ϕ12	19 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATIION-F4



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 2300.000mm
Footing Width - Z (Fw) : 2300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.350m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 125.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.300m

Initial Width (W_o) = 2.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00
38	ULS:(1.2D+1.6(WZ-)+0.5LR	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	401.589	-26.611	-14.586	-4.316	52.583
9	400.532	-26.153	-11.667	1.474	52.974
10	418.586	-28.379	-15.125	-4.504	56.745
11	414.337	-27.937	-14.990	-4.457	55.705
12	402.612	-26.638	-8.838	7.467	52.602
13	400.766	-24.451	-14.638	-4.405	46.311
14	413.740	-14.957	-5.856	7.894	28.553
15	417.511	-17.281	10.281	41.708	42.235
16	240.158	1.340	3.427	13.879	-4.652
17	245.185	-1.758	24.943	58.963	13.590
18	240.130	-13.806	-8.803	-2.679	25.278
19	241.976	-15.994	-3.004	9.193	31.569
20	400.794	-9.305	-2.407	12.153	16.381
21	405.821	-12.403	19.108	57.237	34.623

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	562.225	-37.256	-20.421	-6.042	73.616
23	569.243	-37.498	-16.603	1.970	76.245
24	569.243	-37.498	-16.603	1.970	76.245
25	508.466	-20.917	-8.622	7.695	40.797
26	497.176	-10.084	36.139	93.211	36.445
27	489.133	-5.127	1.714	21.076	7.257
28	497.176	-10.084	36.139	93.211	36.445
29	480.730	-28.847	-17.577	-5.306	54.140
30	483.368	-31.972	-9.292	11.653	63.127
31	360.158	3.740	6.359	22.465	-10.599
32	368.201	-1.216	40.784	94.600	18.589
33	362.891	-23.988	-4.916	12.948	47.352
34	360.254	-20.864	-13.201	-4.011	38.365
35	396.747	-41.561	-49.835	-67.297	68.724
36	236.111	-30.916	-44.000	-65.570	47.691
37	353.683	-47.870	-69.525	-104.654	73.151
38	482.658	-56.737	-74.170	-106.043	91.007

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 5.290m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 2.175m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.300 m Governing Load Case : # 8

Width (W_2) = 2.300 m Governing Load Case : # 8

Depth (D_2) = 0.450 m Governing Load Case : # 38

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 5.290 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 57.130 kN

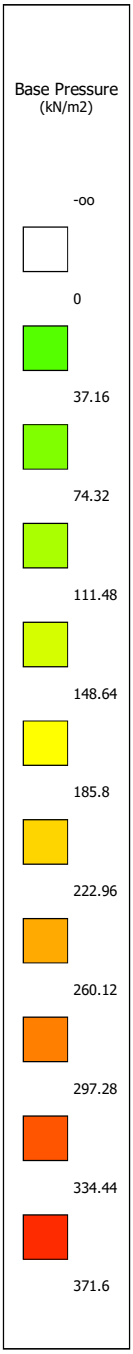
Gross Soil Bearing Capacity = 371.60 kN/m2

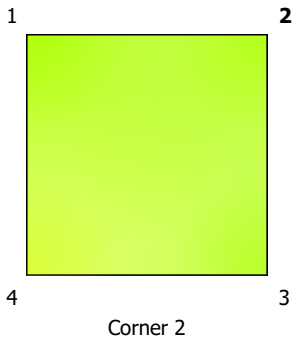
Soil Weight On Top Of Footing = 68.578 kN

Pressures at 4 Corners

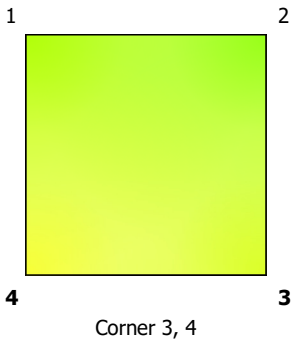
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

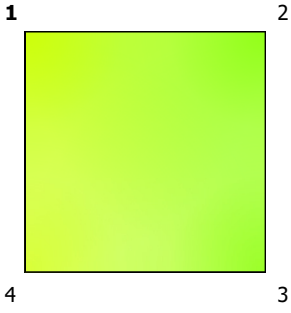




Corner 2

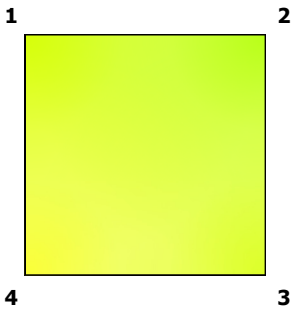


Corner 3, 4



Corner 1

Governing Base Pressure Contour



Corner 1, 2, 3, 4

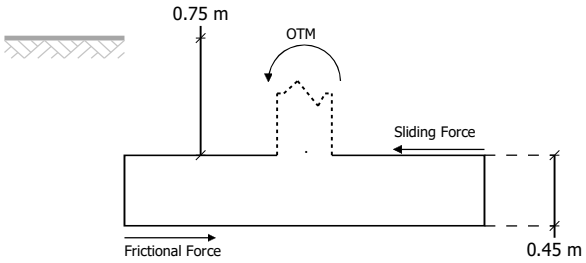
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
10	**142.7500**	74.1872	63.0321	131.5949	0.000
20	104.2119	**83.9259**	94.8435	115.1295	0.000
21	87.8380	48.1847	**113.1179**	152.7712	0.000
21	87.8380	48.1847	113.1179	**152.7712**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
10	**142.7500**	74.1872	63.0321	131.5949
20	104.2119	**83.9259**	94.8435	115.1295
21	87.8380	48.1847	**113.1179**	152.7712
21	87.8380	48.1847	113.1179	**152.7712**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	9.907	18.075	12.287	1.500	55.779	9.429	1.500
9	10.061	22.552	12.994	1.500	160.396	9.383	1.500
10	9.590	17.993	11.968	1.500	55.383	9.037	1.500
11	9.665	18.013	12.044	1.500	55.479	9.130	1.500
12	9.917	29.888	13.311	1.500	174.231	9.442	1.500
13	10.766	17.983	13.063	1.500	55.123	10.604	1.500
14	18.033	46.059	23.747	1.500	118.047	17.647	1.500
15	15.718	26.419	19.103	1.500	13.492	12.537	1.500
16	136.547	53.374	70.302	1.500	27.313	80.500	1.500
17	105.481	7.435	10.488	1.500	6.083	29.819	1.500
18	13.249	20.778	15.798	1.500	63.430	13.433	1.500
19	11.495	61.205	15.977	1.500	53.981	10.967	1.500
20	28.292	109.361	38.736	1.500	54.740	29.550	1.500
21	21.428	13.908	16.499	1.500	9.292	15.261	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 10
Governing Disturbing Force : -28.379kN
Governing Restoring Force : 272.147kN
Minimum Sliding Ratio for the Critical Load Case : 9.590
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 70.187kNm
Governing Resisting Moment : 426.984kNm
Minimum Overturning Ratio for the Critical Load Case : 6.083

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 24.943kN

Governing Restoring Force : 185.446kN

Minimum Sliding Ratio for the Critical Load Case : 7.435

Critical Load Case for Overturning about Z-Direction : 10

Governing Overturning Moment : 69.516kNm

Governing Resisting Moment : 628.235kNm

Minimum Overturning Ratio for the Critical Load Case : 9.037

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 25.005kN

Governing Restoring Force : 262.261kN

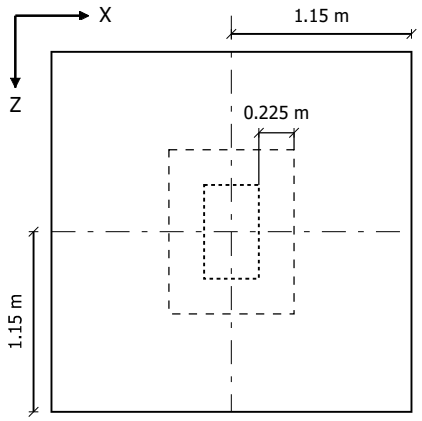
Minimum Sliding Ratio for the Critical Load Case : 10.488

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 1.714

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 597.797kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.476m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2915.357kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4395.874kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2691.098kN

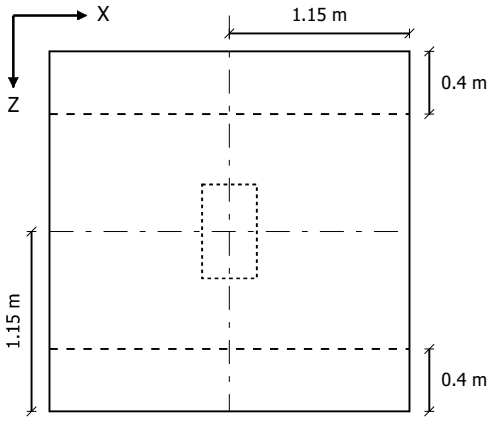
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 2018.324kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 890.323kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.456m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

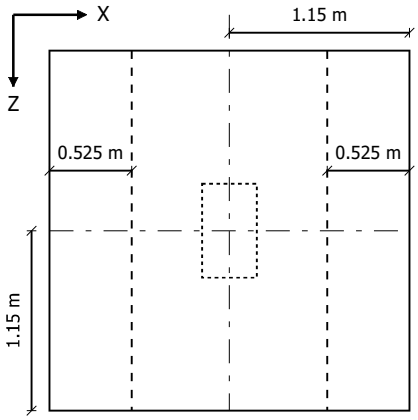
From above calculations, 0.75 X V_c = 667.742 kN

Critical load case for V_{ux} is # 38 $V_{ux} = V_{ux}|_{x=D_x}$ = 178.428 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 890.323 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.581 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

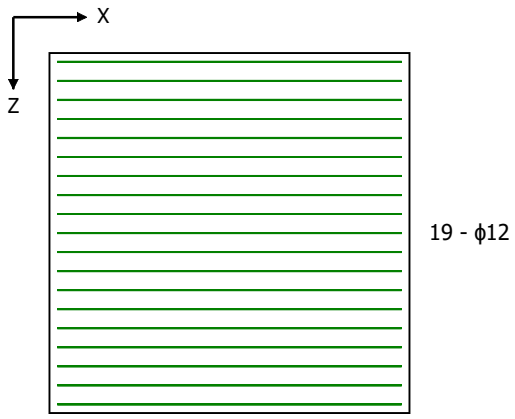
From above calculations, 0.75 X V_c = 667.742 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 221.414 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00165	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.975	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	179.617	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	199.574	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00134	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00118	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1703.175	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 125.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 125mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.925 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 19

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 2148.834 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.394 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

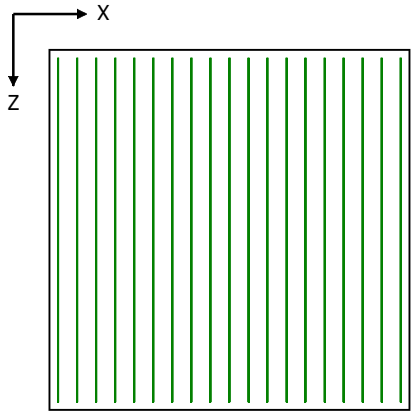
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.262

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



19 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 38

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.382 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00161

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.850 m

Ultimate moment, $M_u|_{x=D_x}$ = 138.591 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 153.990 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00110

(Based on gross depth) ρ x d_{eff} / Depth = 0.00093

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1665.087 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 125.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 125mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.800 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 19

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 2148.834 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.382 m

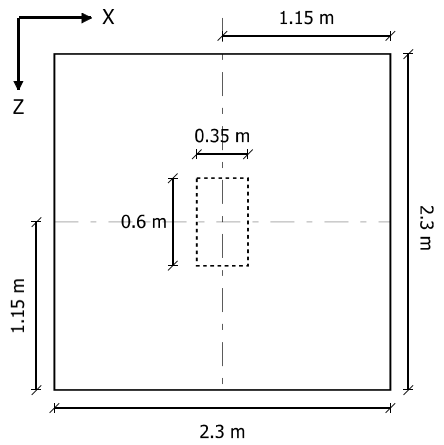
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00245

From ACI Cl.7.6.1, minimum req'd clear distance between bars

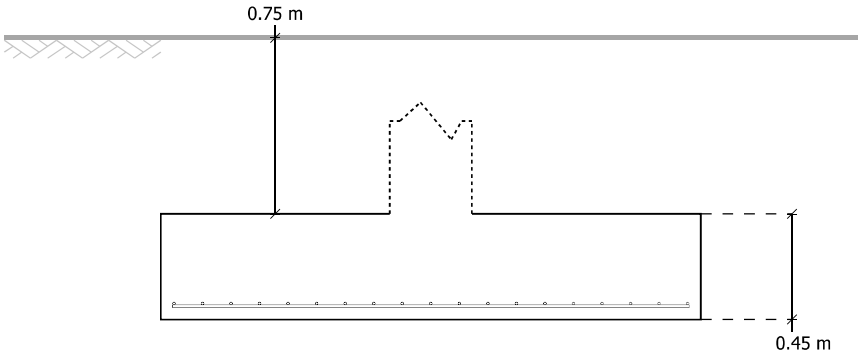
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.291

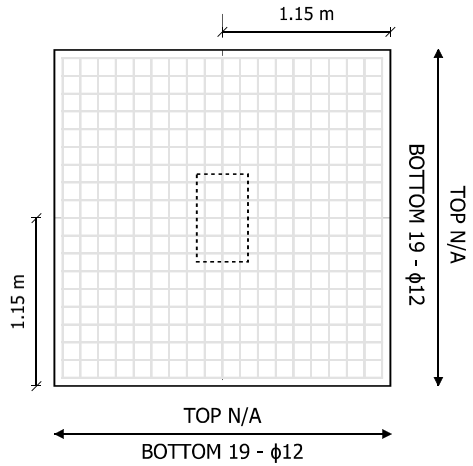
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.3 m	2.3 m	0.45 m	N/A	N/A	19 - ϕ12	19 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

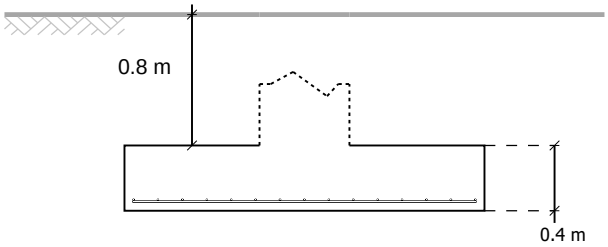
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

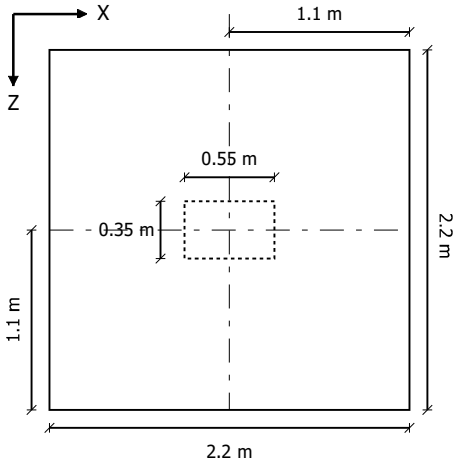
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
95	1	2.200m	2.200m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
95	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATIION-F5



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2200.000mm
Footing Width - Z (Fw) : 2200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.200m

Initial Width (W_o) = 2.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00
38	ULS:(1.2D+1.6(WZ-)+0.5LR	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	282.278	-5.602	-0.064	5.017	15.340
9	281.296	-8.759	-0.082	5.222	24.528
10	293.700	-7.130	-0.068	5.529	16.691
11	290.844	-6.748	-0.067	5.401	16.353
12	282.278	-5.610	1.130	9.367	15.353
13	282.000	-3.569	-0.064	5.008	11.236
14	290.176	-0.905	23.838	69.676	5.743
15	290.803	-6.998	19.975	59.589	16.593
16	168.476	4.429	31.835	88.710	-4.943
17	169.311	-3.695	26.685	75.261	9.524
18	169.089	-1.328	-0.038	3.001	5.100
19	169.367	-3.370	1.155	7.360	9.217
20	281.388	2.189	31.809	90.717	1.193
21	282.222	-5.936	26.659	77.268	15.660

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	395.189	-7.843	-0.089	7.024	21.476
23	399.525	-13.026	-0.116	7.567	35.015
24	399.525	-13.026	-0.116	7.567	35.015
25	356.296	-2.935	25.415	75.399	9.252
26	344.355	-8.020	42.678	121.878	19.596
27	343.020	4.979	50.918	143.396	-3.552
28	344.355	-8.020	42.678	121.878	19.596
29	338.336	-3.818	-0.076	6.008	12.545
30	338.734	-6.734	1.628	12.235	18.427
31	252.625	7.423	50.939	141.635	-8.829
32	253.961	-5.576	42.699	120.117	14.318
33	254.050	-5.054	1.647	10.730	13.825
34	253.653	-2.137	-0.057	4.503	7.943
35	282.508	-9.909	-33.229	-84.642	20.651
36	169.597	-7.668	-33.203	-86.648	14.515
37	254.419	-11.932	-53.121	-138.938	22.304
38	344.813	-14.377	-53.142	-137.177	27.581

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.840m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.842m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.200 m Governing Load Case : # 8

Width (W_2) = 2.200 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.840 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 46.462 kN

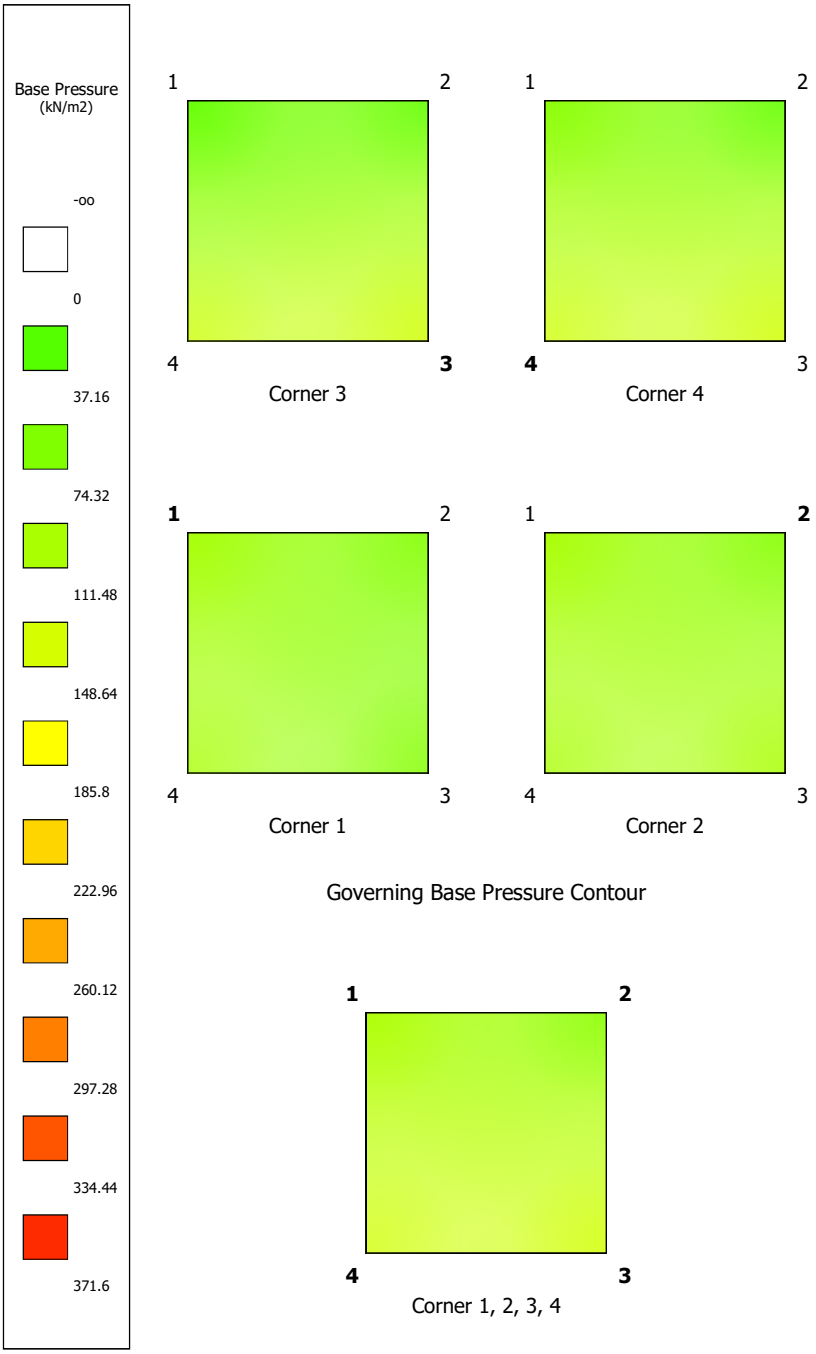
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 66.922 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



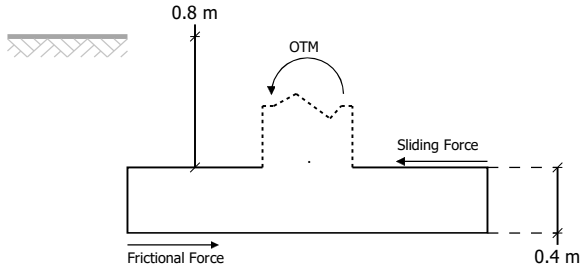
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**94.4170**	62.8257	68.6739	100.2653	0.000
13	86.0191	**71.7472**	77.3627	91.6345	0.000
20	23.4553	23.0974	**139.6735**	140.0313	0.000
21	42.3498	22.0255	121.1239	**141.4482**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**94.4170**	62.8257	68.6739	100.2653
13	86.0191	**71.7472**	77.3627	91.6345
20	23.4553	23.0974	**139.6735**	140.0313
21	42.3498	22.0255	121.1239	**141.4482**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	35.315	3096.820	49.939	1.500	87.610	24.793	1.500
9	22.531	2420.192	31.862	1.500	84.062	15.511	1.500
10	28.547	2984.661	40.369	1.500	81.768	22.947	1.500
11	29.951	3011.371	42.355	1.500	83.125	23.373	1.500
12	35.261	175.137	48.886	1.500	44.537	24.770	1.500
13	55.393	3105.579	78.325	1.500	87.703	34.395	1.500
14	222.920	8.465	11.962	1.500	5.630	72.819	1.500
15	28.877	10.117	13.503	1.500	6.610	22.960	1.500
16	31.817	4.427	6.201	1.500	3.077	46.271	1.500
17	38.256	5.297	7.420	1.500	3.643	28.323	1.500
18	106.345	3706.599	150.333	1.500	104.750	55.291	1.500
19	41.955	122.389	56.127	1.500	40.027	29.501	1.500
20	90.186	6.205	8.755	1.500	4.218	1369.584	1.500
21	33.325	7.420	10.242	1.500	4.972	24.166	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 9
Governing Disturbing Force : -8.759kN
Governing Restoring Force : 197.340kN
Minimum Sliding Ratio for the Critical Load Case : 22.531
Critical Load Case for Overturning about X-Direction : 16
Governing Overturning Moment : 101.443kNm
Governing Resisting Moment : 312.120kNm
Minimum Overturning Ratio for the Critical Load Case : 3.077

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 16

Governing Disturbing Force : 31.835kN

Governing Restoring Force : 140.930kN

Minimum Sliding Ratio for the Critical Load Case : 4.427

Critical Load Case for Overturning about Z-Direction : 9

Governing Overturning Moment : 28.032kNm

Governing Resisting Moment : 434.793kNm

Minimum Overturning Ratio for the Critical Load Case : 15.511

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 16

Governing Disturbing Force : 32.141kN

Governing Restoring Force : 199.305kN

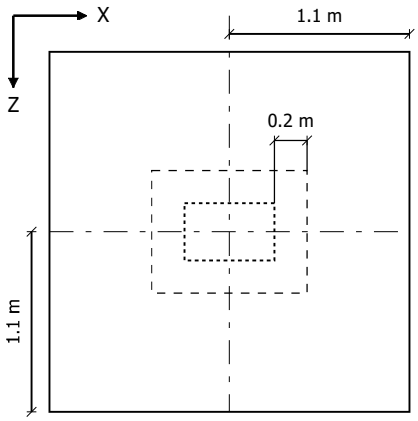
Minimum Sliding Ratio for the Critical Load Case : 6.201

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.571

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 447.160kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2439.551kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

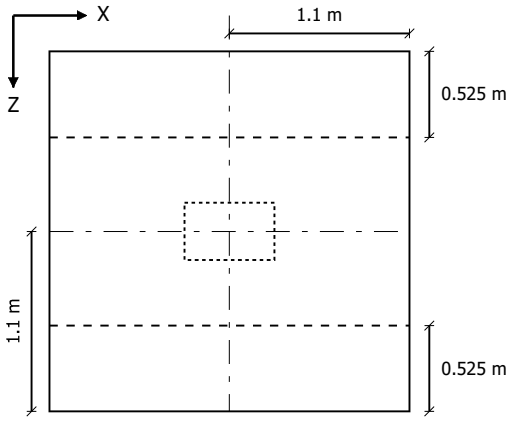
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 743.541kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.581m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

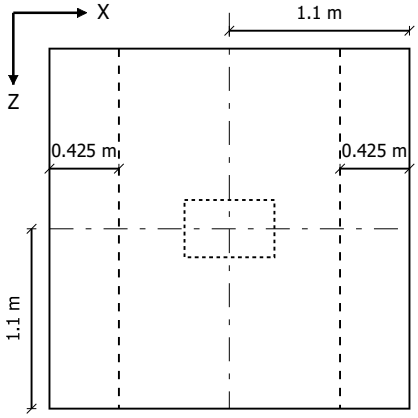
From above calculations, 0.75 X V_c = 557.655 kN

Critical load case for V_{ux} is # 27 $V_{ux} = V_{ux}|_{x=D_x}$ = 207.334 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 743.541 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.481 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

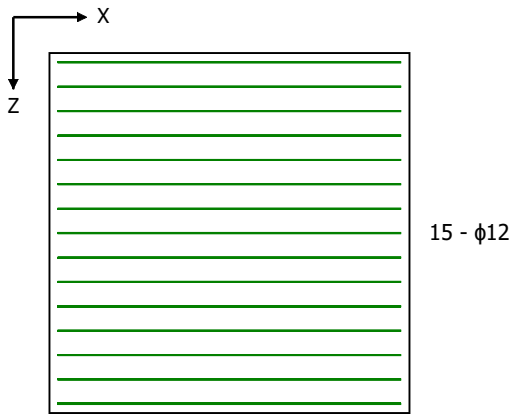
From above calculations, 0.75 X V_c = 557.655 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 130.882 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00156	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.825	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	92.063	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	102.293	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00094	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00081	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1374.430	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover} =$ 0.775 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar} = \frac{A_s}{A_{bar}} =$ 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho = \frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00224

From ACI Cl.7.6.1, minimum req'd clear distance between bars

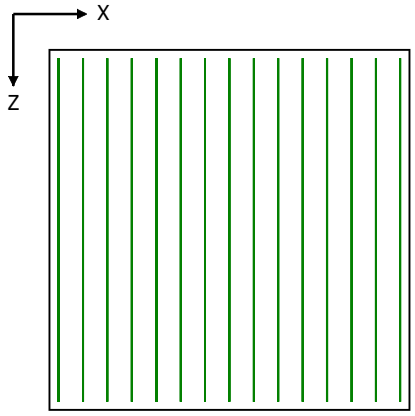
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.234

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal} = 0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max} = 0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00162

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.275 m

Ultimate moment, $M_u|_{x=D_x}$ = 151.308 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 168.120 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00167

(Based on gross depth) ρ x d_{eff} / Depth = 0.00139

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1428.597 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.875 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

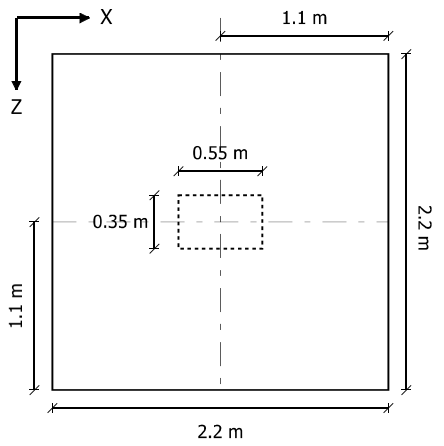
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

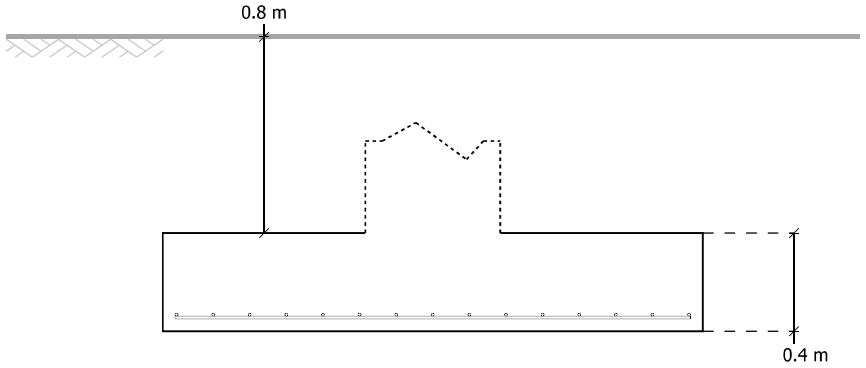
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.187

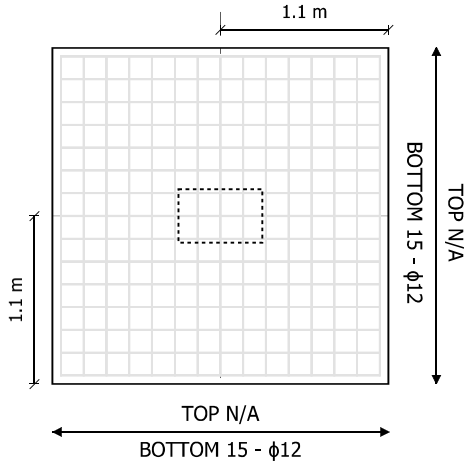
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.2 m	2.2 m	0.4 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

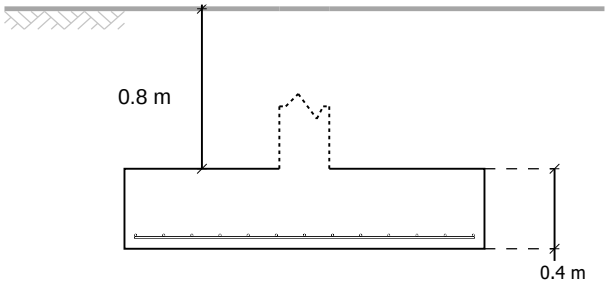
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

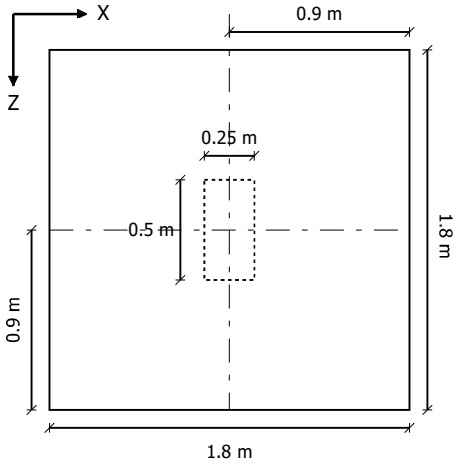
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
77	1	1.800m	1.800m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
77	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATIION-F6



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1800.000mm
Footing Width - Z (Fw) : 1800.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 150.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.800m

Initial Width (W_o) = 1.800m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00
38	ULS:(1.2D+1.6(WZ-)+0.5LR	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	214.721	-4.708	-25.700	-10.004	6.651
9	216.583	-7.163	-28.445	-10.137	12.360
10	220.508	-4.859	-25.734	-10.029	7.402
11	219.061	-4.821	-25.726	-10.023	7.214
12	214.988	-4.736	-24.314	-7.597	6.710
13	213.573	-3.536	-25.690	-9.990	4.312
14	215.642	9.864	-25.421	-9.767	-8.916
15	220.357	-16.710	-20.583	-1.588	18.338
16	124.274	16.755	-15.014	-5.661	-17.517
17	130.561	-18.676	-8.563	5.244	18.821
18	127.685	-1.653	-15.411	-5.988	1.652
19	129.099	-2.853	-14.034	-3.596	4.050
20	210.162	14.872	-25.294	-9.663	-14.857
21	216.449	-20.559	-18.843	1.243	21.482

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	300.609	-6.591	-35.979	-14.005	9.311
23	306.109	-10.104	-39.840	-14.204	17.680
24	306.109	-10.104	-39.840	-14.204	17.680
25	263.277	9.773	-30.570	-11.773	-8.022
26	263.323	-31.087	-19.886	5.977	32.086
27	253.264	25.603	-30.207	-11.472	-26.055
28	263.323	-31.087	-19.886	5.977	32.086
29	256.026	-3.976	-30.827	-11.985	4.640
30	258.046	-5.690	-28.860	-8.567	8.066
31	185.954	27.091	-22.480	-8.458	-28.426
32	196.014	-29.599	-12.159	8.991	29.715
33	193.630	-4.278	-21.150	-5.566	6.071
34	191.609	-2.563	-23.117	-8.984	2.645
35	215.059	-20.700	-32.636	-21.331	21.375
36	129.171	-18.817	-22.356	-17.330	18.715
37	193.790	-29.825	-34.228	-27.127	29.545
38	261.100	-31.313	-41.955	-30.141	31.916

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.240m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.134m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.800 m Governing Load Case : # 8

Width (W_2) = 1.800 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 38

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.240 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 31.103 kN

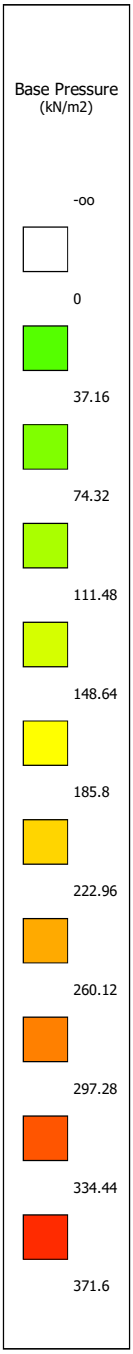
Gross Soil Bearing Capacity = 371.60 kN/m2

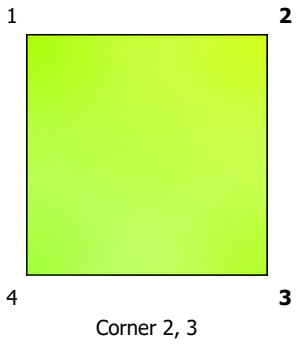
Soil Weight On Top Of Footing = 44.854 kN

Pressures at 4 Corners

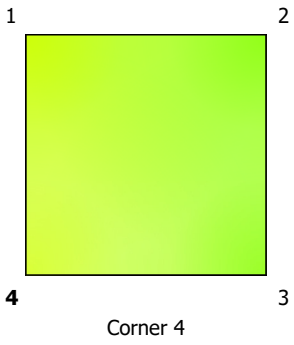
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

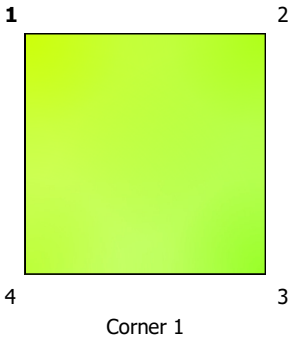




Corner 2, 3

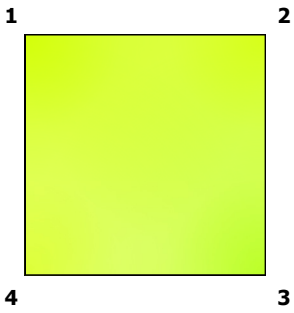


Corner 4



Corner 1

Governing Base Pressure Contour



Corner 1, 2, 3, 4

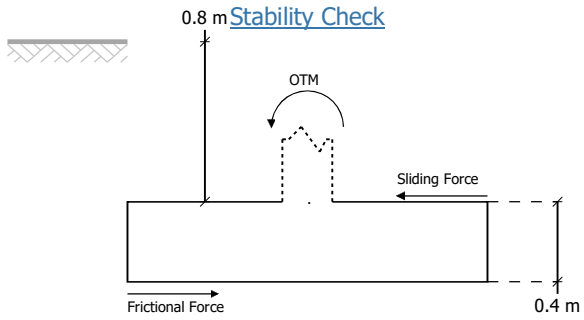
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**128.0894**	96.7602	52.4908	83.8199	0.000
20	87.2534	**130.0635**	89.3634	46.5533	0.000
20	87.2534	130.0635	**89.3634**	46.5533	0.000
21	127.2860	66.1634	53.2117	**114.3343**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**128.0894**	96.7602	52.4908	83.8199
20	87.2534	**130.0635**	89.3634	46.5533
20	87.2534	130.0635	**89.3634**	46.5533
21	127.2860	66.1634	53.2117	**114.3343**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	30.871	5.655	7.867	1.500	12.898	30.798	1.500
9	20.420	5.142	7.052	1.500	12.237	17.372	1.500
10	30.507	5.760	8.005	1.500	13.129	28.679	1.500
11	30.595	5.734	7.970	1.500	13.071	29.173	1.500
12	30.714	5.983	8.305	1.500	15.116	30.571	1.500
13	40.938	5.635	7.895	1.500	12.858	45.716	1.500
14	14.781	5.735	7.562	1.500	13.164	20.499	1.500
15	8.867	7.198	7.903	1.500	27.153	10.707	1.500
16	5.975	6.668	6.293	1.500	15.446	7.491	1.500
17	5.529	12.059	7.108	1.500	102.185	7.115	1.500
18	61.596	6.607	9.291	1.500	15.081	79.763	1.500
19	35.936	7.306	10.125	1.500	20.039	35.782	1.500
20	9.619	5.656	6.895	1.500	13.018	12.435	1.500
21	7.111	7.759	7.414	1.500	41.809	8.900	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 17
Governing Disturbing Force : -18.676kN
Governing Restoring Force : 103.259kN
Minimum Sliding Ratio for the Critical Load Case : 5.529
Critical Load Case for Overturning about X-Direction : 9
Governing Overturning Moment : -21.515kNm
Governing Resisting Moment : 263.281kNm
Minimum Overturning Ratio for the Critical Load Case : 12.237

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 9

Governing Disturbing Force : -28.445kN

Governing Restoring Force : 146.270kN

Minimum Sliding Ratio for the Critical Load Case : 5.142

Critical Load Case for Overturning about Z-Direction : 17

Governing Overturning Moment : 26.292kNm

Governing Resisting Moment : 187.078kNm

Minimum Overturning Ratio for the Critical Load Case : 7.115

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 16

Governing Disturbing Force : 22.498kN

Governing Restoring Force : 141.585kN

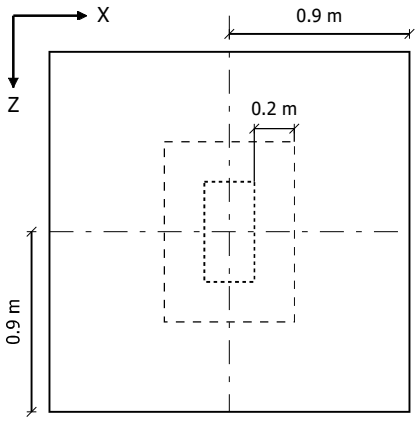
Minimum Sliding Ratio for the Critical Load Case : 6.293

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 322.948kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $\left(2 + \frac{4}{\beta_c}\right) \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 1944.021kN

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d$ = 3297.265kN

Equation 11-33, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 1944.021kN

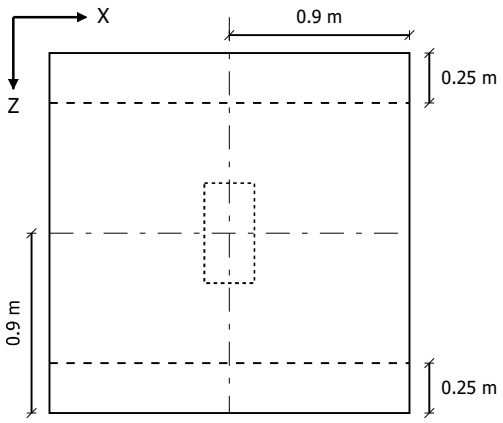
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.306m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

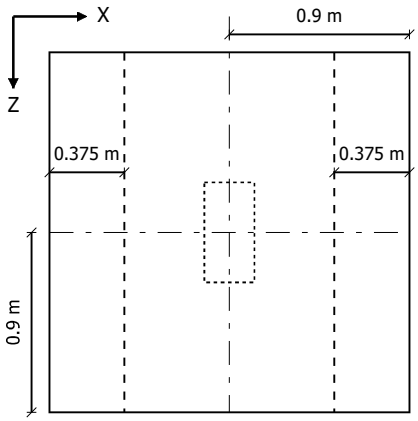
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{ux} is # 38 $V_{ux} = V_{ux}|_{x=D_x}$ = 79.369 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

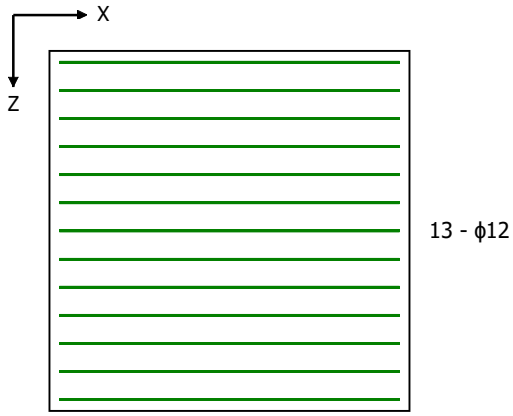
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{uz} is # 26 $V_{uz} = V_{uz}|_{z=D_z}$ = 108.265 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times F_c' \times \frac{87}{\left[f_y \times \left(87 + F_y\right)\right]}$ =	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00168
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.775 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	74.252 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	82.502 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00093
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00080
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1211.405 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 150.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.725	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	13	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1470.255	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00237	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

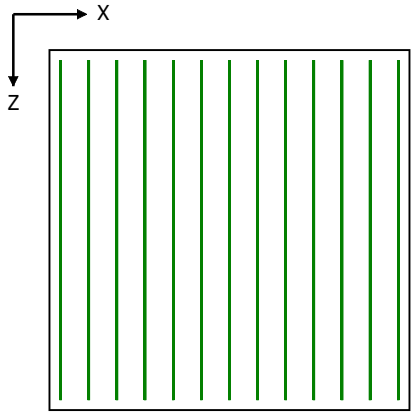
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.214

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times F_c' \times \frac{87}{[f_y \times (87 + F_y)]}$ =	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00164	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	0.650	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	53.789	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	59.766	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00072	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00060	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1183.814	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 150.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times \left(L - D_{col}\right) - C_{cover}$ =	0.600	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	13	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

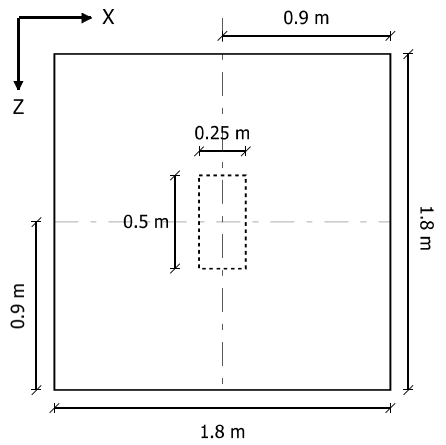
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1470.255	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.332	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ =	0.00246	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

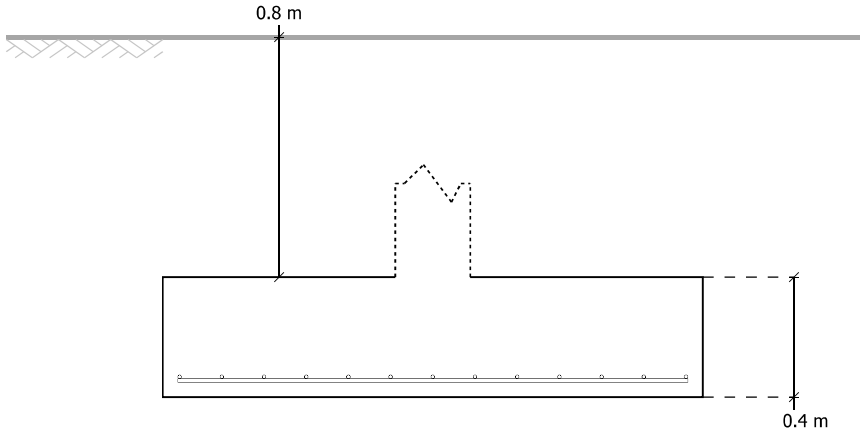
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.242

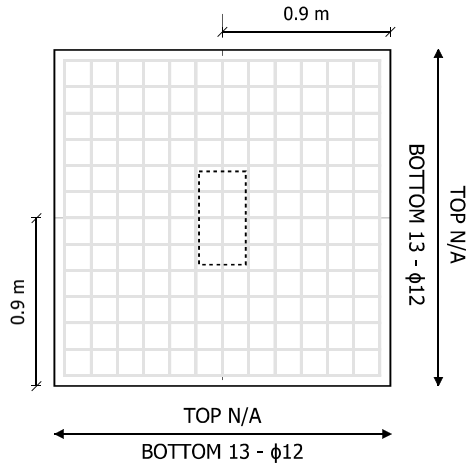
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.8 m	1.8 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet